College Physics 2

Electric Current and DC Circuits

MPS Department | FEU Institute of Technology

College Physics 2

Current, Resistance and Ohm's Law

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OBJECTIVES

- Discuss the concepts of electric current, drift velocity and current density
- Differentiate the resistivity of conductor and insulators
- State Ohm's law and solve problems involving Ohm's law

CURRENT

- **Motion of charge** from one region to another
- In electrostatics, electric field everywhere within a conductor is zero
  - **no current**
- In general, free electrons move randomly in a conductor
- But since the movement is random there is no net flow in any direction → **no current**

CURRENT

Electric current is the rate of flow of electrons in a circuit.

The unit of current, called ampere, is named in honor of Andre' Marie Ampere (1775-1836), a French physicist. It is expressed in terms of the number of coulombs per second going past a given point on a wire.

1 A = 1 Coulomb/second = 1 C/s

$$Current, I = \frac{charge}{time} = \frac{q}{t}$$

> **The flow of charge in a wire is called current.**

CURRENT

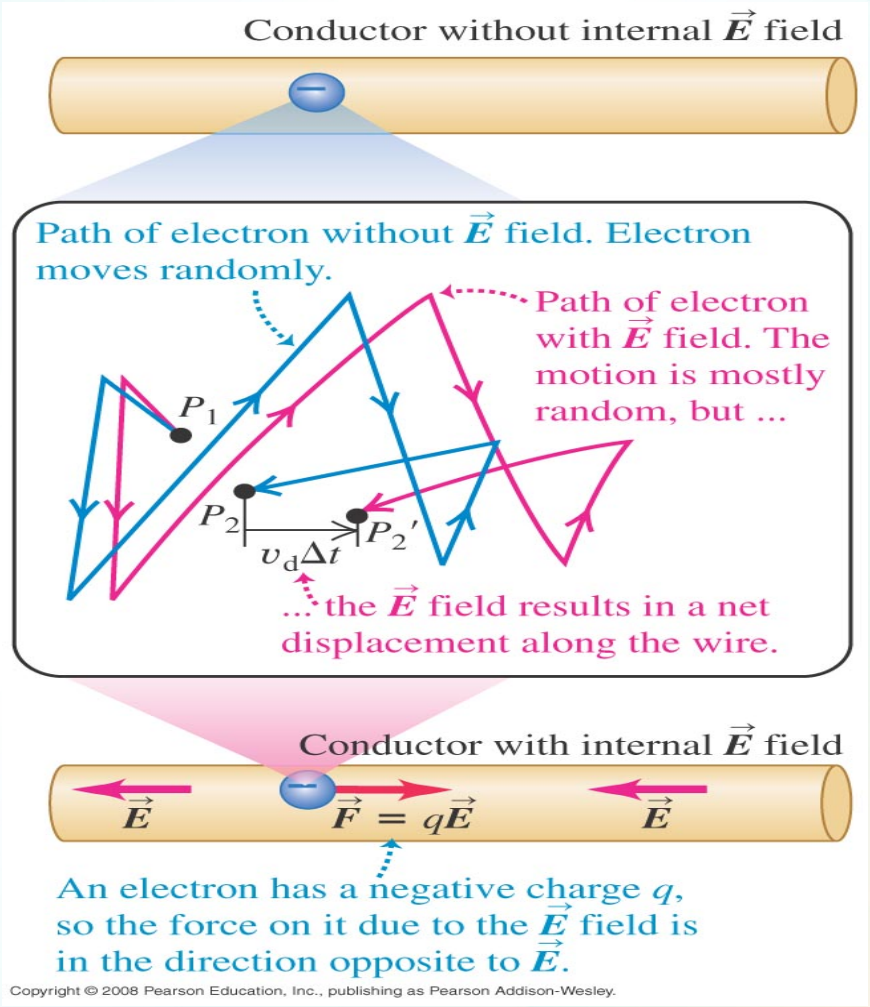
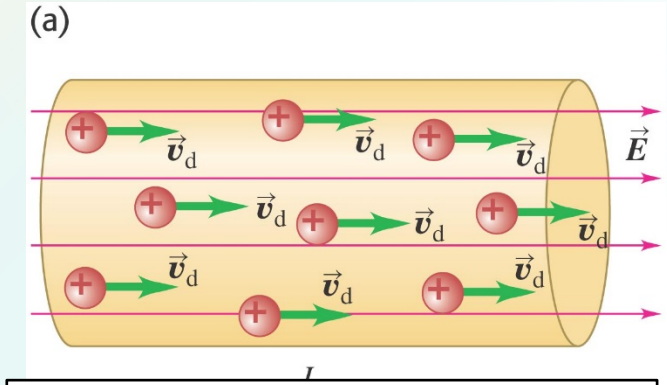


Figure 1. Random motion of the charged particles.. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 819) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

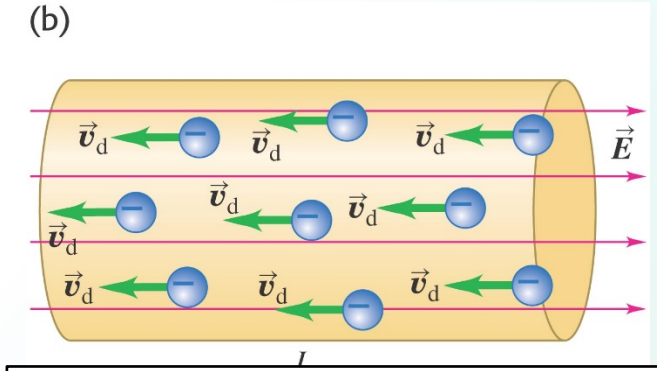
Direction of current

- v_d – result of acceleration of the electrons due to the electric field and the repeated collisions with the atoms.

$$v_d = \vec{a}\ due\ to\ \vec{E} \times collision\ time\ \tau_{coll}$$



(+) charges: v_d **same** direction as e-field

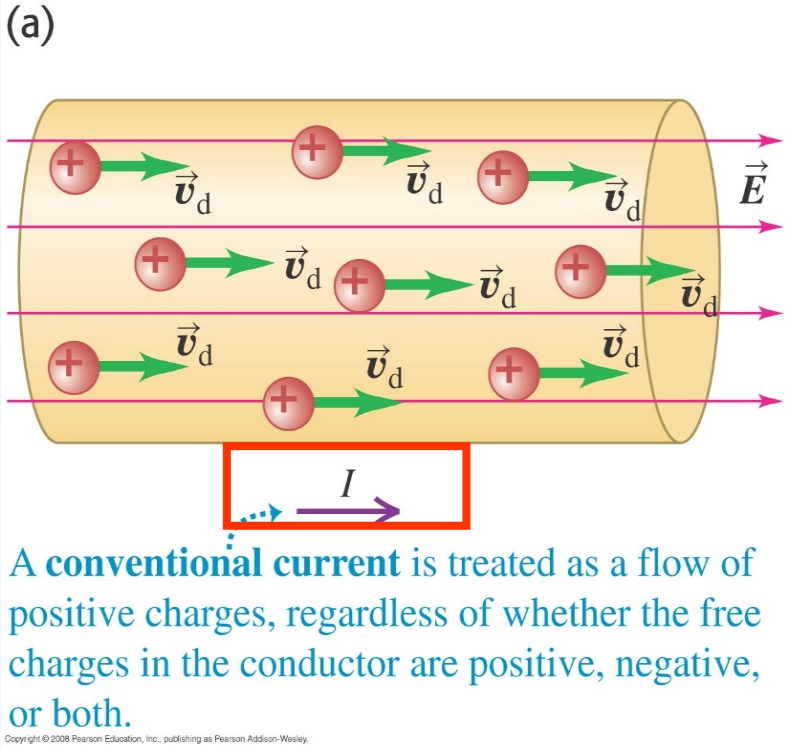


(-) charges: v_d **opposite** direction as e-field

Figure 2. Current produce by a) a positive charge and b) a negative charge.. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 820) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

Direction of current

- Direction of *I* is in the same direction where there is a flow of positive charge – **conventional current**



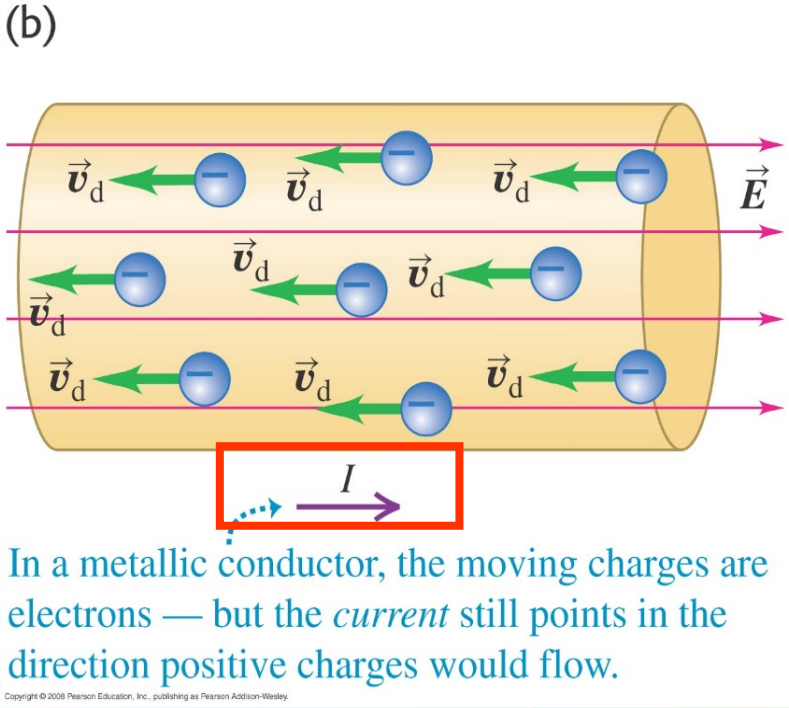
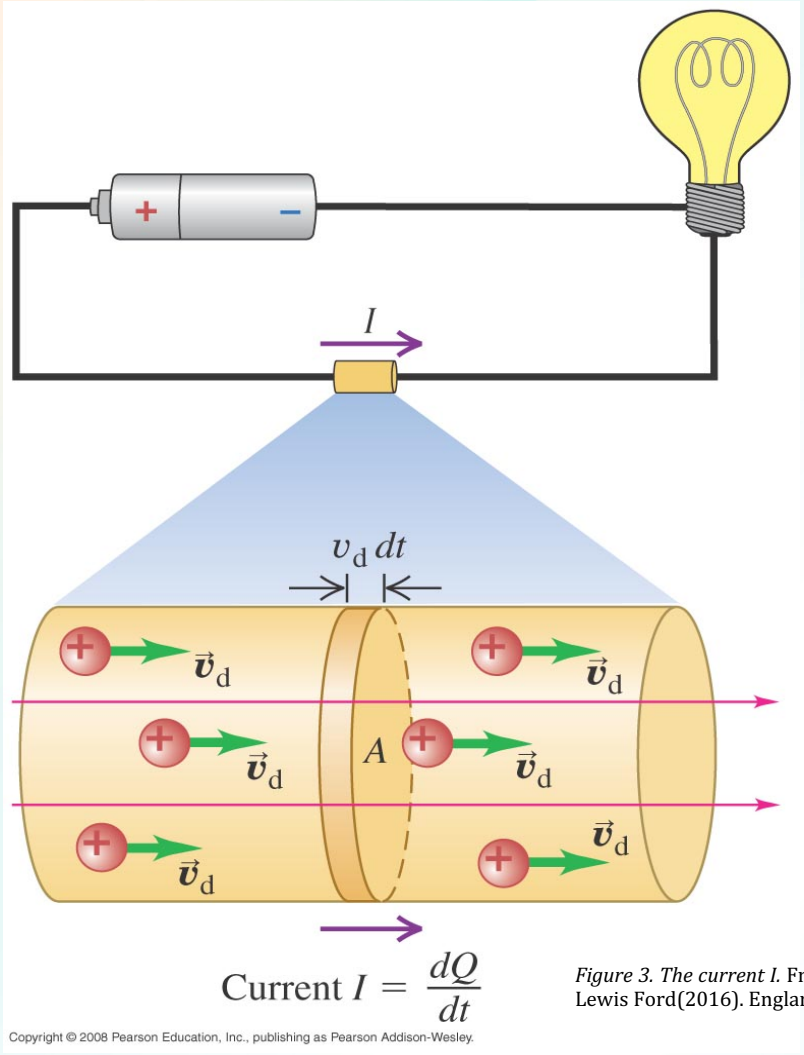


Figure 2. Current produce by a) a positive charge and b) a negative charge.. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 820) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

Definition of Current



- Net charge flowing through the area per unit time

$$I = \frac{dQ}{dt}$$

- Current is not a vector
- **SI Unit:** Ampere (A) or (C/s)

Figure 3. The current I. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 825) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

CURRENT DENSITY

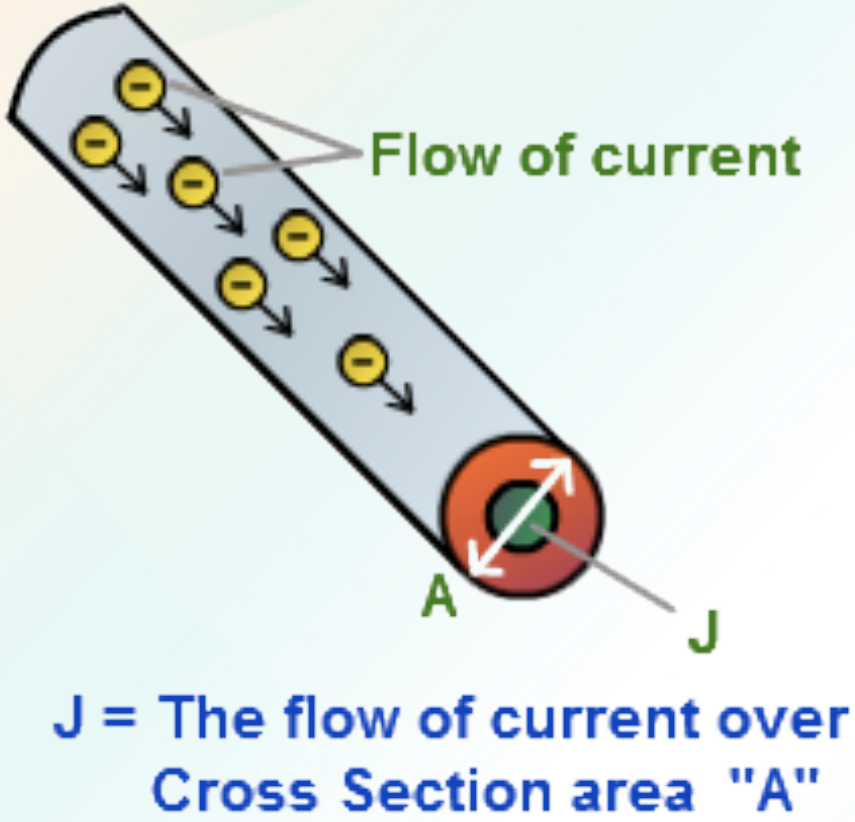


Figure 4. Current Density. *http://s3-ap-southeast-1.amazonaws.com/subscriber.images/physics/2016/10/27125734/Current-Density.png*

> The amount of electric current flowing per unit cross-sectional area of a material.

DRIFT VELOCITY

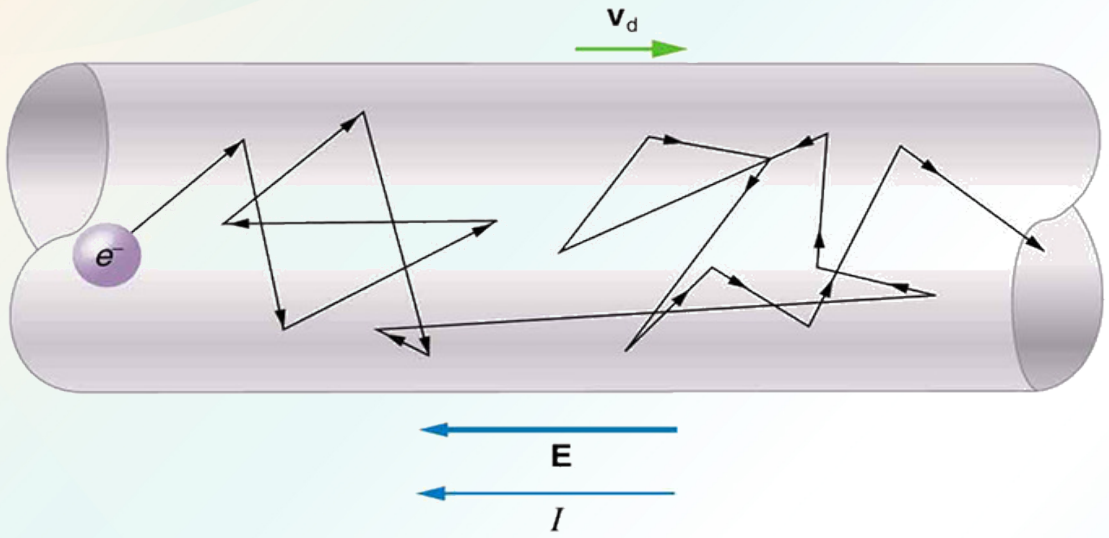


Figure 5. Drift Velocity. From. *https://tse1.mm.bing.net/th?id=OIP.sFSTlowlBvCRq9CZ9O0LIQHaD5&pid=Api&P=0&w=344&h=182*

The direction of drift velocity is opposite to the electric field for electrons.

The electric field in the conductor causes the electrons to drift in the direction opposite to the field, since they are negative.

The average velocity of the free charges is called the drift velocity

Current, Drift Velocity and Current Density

- **Current density** – current per cross sectional area

$$J = \frac{I}{A} = nqv_d$$

- **Units:** A/m^2
- If the charges are negative, it moves opposite to **E** but the current **is still** along **E**
- *I* and *J* don't depend on the sign of the charge

Current, Drift Velocity and Current Density

$$I = \frac{dQ}{dt} = n|q|v_D A$$

$$J = \frac{I}{A} = n|q|v_D$$

$$\vec{J} = nq\vec{v}_D$$

Vector current density is **always** in the direction of the field

Current and Current Density

- Current due to an ensemble of charges is the **sum of the individual currents** for each charge
- Similarly, total current density is just the **sum of the individual current densities**

Effect of Electric Current on Human Body

Current in milli amperes	Effects
1 or Less	No sensation; probably not notice
1 to 3	Mild sensation not paintful
3 to 10	Muscular control could be lost or muscle clamping
30 to 75	Respiratory paralysis
75mA to 4 Amps	Ventricular Fibrillation
Over 4 Amps	Tissue begins to burns. Heart muscles clamp and heart stops beating

Table 1. Effect of electric current of human body. From. *http://2.bp.blogspot.com/-oV6vCLqwnc8/TpFltQmBw9I/AAAAAAAAAEY/qHdnVQwSfI4/s1600/Current+Effects.JPG*

Sample Problem

The current density J in a wire is given by $50.0\text{A/m}^2\,i$. If the drift velocity of the charge carriers (q = -0.1C) is $v_d = -2.0 \times 10^{-4}\text{m/s}\,i$, what is the concentration of the carriers?

$$J = n|q|v_d \quad \rightarrow n = \frac{J}{|q|v_d}$$

$$n = \frac{50.0\,A/m^2}{|-0.1C|2.0\times10^{-4}\,m/s} = 2.5\times10^{6}\,m^{-3}$$

Sample Problem

1. How much current is flowing when 29.3 C of charge pass a point in 7.84 seconds?

Answer: I = 3.74 A

2. A calculator draws a current of 0.01 A for 60 seconds. Calculate the magnitude of charge.

Answer: Q = 0.6 C

3. 15 C of charge flow past a given point in a wire in 3 seconds. What is the current in Amperes?

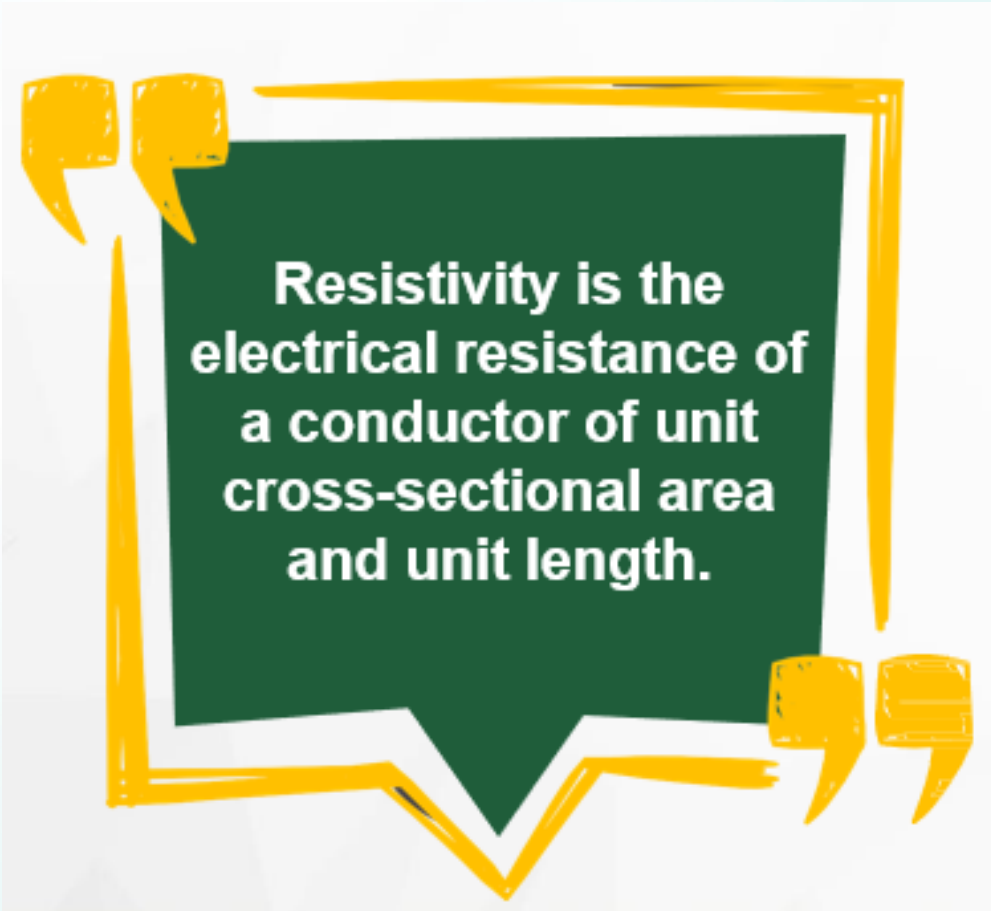
Answer: I = 5 A

RESISTIVITY

Resistivity is a characteristic property of each material.

It is useful in comparing various materials on the basis of their ability to conduct electric currents.

High resistivity designates poor conductors.



RESISTIVITY

Resistivity, is commonly symbolized by the Greek letter rho, ρ, is quantitatively equal to the resistance, R, of a specimen such as a wire, multiplied by its cross-sectional area, A, and divided by its length, l.

$$\rho = \frac{RA}{\ell}$$

Where

ρ = Resistivity, ohm-meter (Ωm)
R = resistance, Ohm (Ω)
A = wire cross-sectional area (m^2)
l = length (m)

GEORG SIMON OHM
German physicist (1789–1854)

Figure 5. Georg Simon Ohm. From. *Physics for Scientists and Engineers with Modern Physics*(pp 814) by Serway, R. A., & Jewett, J. W. (2019). Australia: Cengage Learning.

RESISTIVITY

$$\Delta R = \alpha R \Delta T$$

The value of resistivity depends also on the temperature of the material.

Where:

ΔR – change in resistance, Ω
R - resistance of the substance, Ω
ΔT – change in temperature, ºC
α – temperature coefficient of resistance of the material (1/ºC)

Resistivities and Temperature Coefficients of Resistivity for Various Materials

Material	Resistivity[a] ($\Omega \cdot m$)	Temperature Coefficient[b] α $[(°C)^{-1}]$
Silver	1.59×10^{-8}	3.8×10^{-3}
Copper	1.7×10^{-8}	3.9×10^{-3}
Gold	2.44×10^{-8}	3.4×10^{-3}
Aluminum	2.82×10^{-8}	3.9×10^{-3}
Tungsten	5.6×10^{-8}	4.5×10^{-3}
Iron	10×10^{-8}	5.0×10^{-3}
Platinum	11×10^{-8}	3.92×10^{-3}
Lead	22×10^{-8}	3.9×10^{-3}
Nichrome[c]	1.00×10^{-6}	0.4×10^{-3}
Carbon	3.5×10^{-5}	-0.5×10^{-3}
Germanium	0.46	-48×10^{-3}
Silicon[d]	2.3×10^{3}	-75×10^{-3}
Glass	10^{10} to 10^{14}	
Hard rubber	$\sim 10^{13}$	
Sulfur	10^{15}	
Quartz (fused)	75×10^{16}	

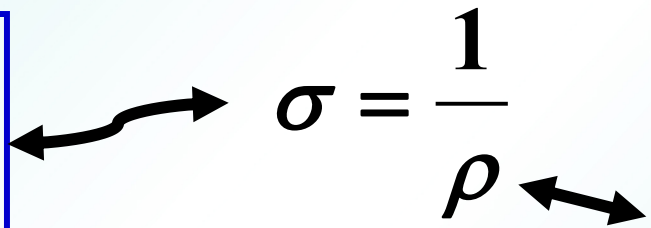
[a] All values at 20°C. All elements in this table are assumed to be free of impurities.
[b] See Section 27.4.
[c] A nickel–chromium alloy commonly used in heating elements. The resistivity of Nichrome varies with composition and ranges between 1.00×10^{-6} and $1.50 \times 10^{-6}\ \Omega \cdot m$.
[d] The resistivity of silicon is very sensitive to purity. The value can be changed by several orders of magnitude when it is doped with other atoms.

Table 2. Resistivities and Temperature Coefficients. From. *Physics for Scientists and Engineers with Modern Physics*(pp 814) by Serway, R. A., & Jewett, J. W. (2019). Australia: Cengage Learning.

CONDUCTIVITY

- Inverse of resistivity

How much the material permits flow of current

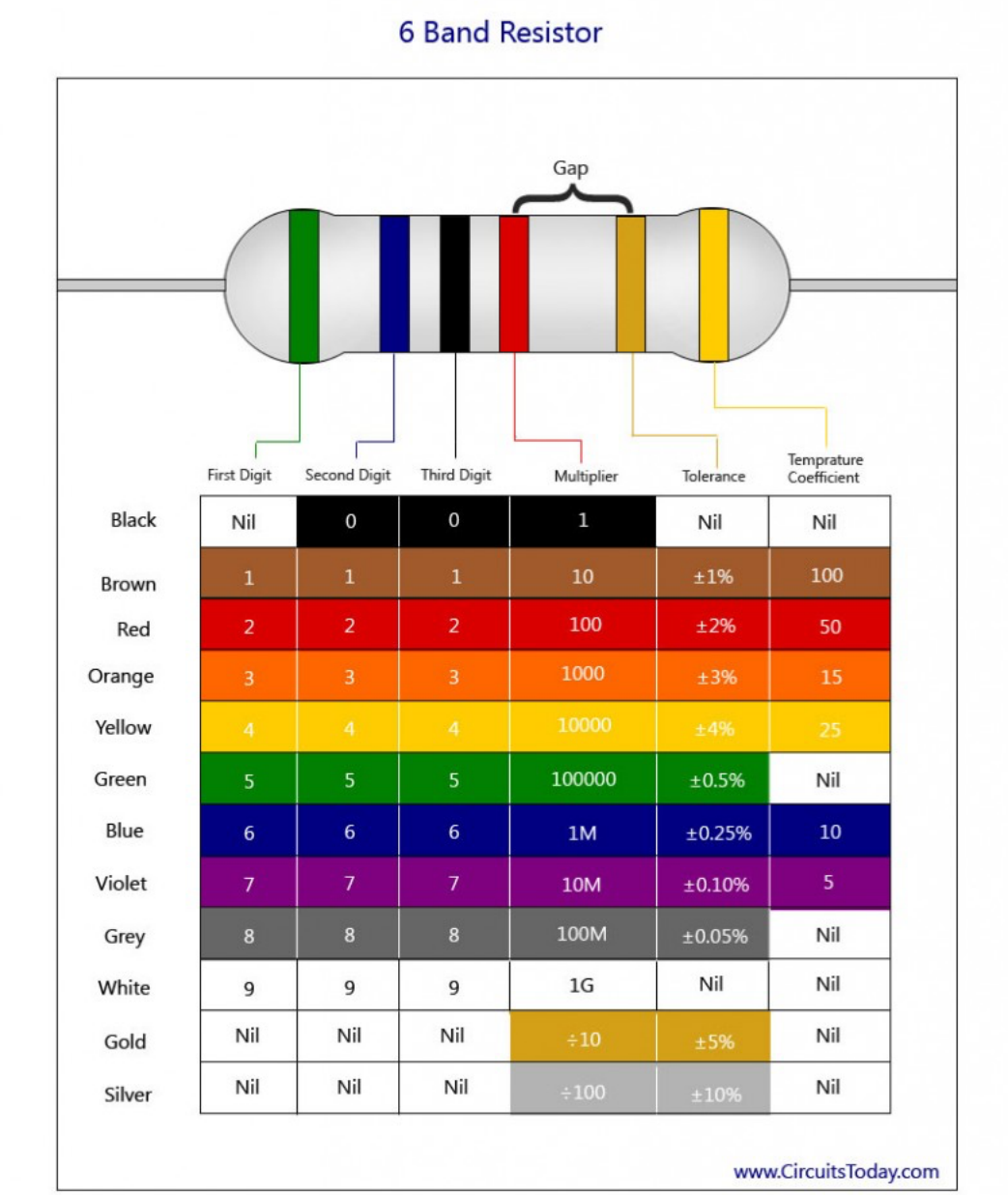


How much the material resists flow of current

- Good conductor: ↑σ, ↓ρ
- Good insulator: ↓σ, ↑ρ
- Conductivity is the electric analogy of thermal conductivity

RESISTORS

Resistor – is a passive electrical component that provides resistance in the electric current in the circuit. Or in other words, it limits the current in the circuit.



	First Digit	Second Digit	Third Digit	Multiplier	Tolerance	Temprature Coefficient
Black	Nil	0	0	1	Nil	Nil
Brown	1	1	1	10	±1%	100
Red	2	2	2	100	±2%	50
Orange	3	3	3	1000	±3%	15
Yellow	4	4	4	10000	±4%	25
Green	5	5	5	100000	±0.5%	Nil
Blue	6	6	6	1M	±0.25%	10
Violet	7	7	7	10M	±0.10%	5
Grey	8	8	8	100M	±0.05%	Nil
White	9	9	9	1G	Nil	Nil
Gold	Nil	Nil	Nil	÷10	±5%	Nil
Silver	Nil	Nil	Nil	÷100	±10%	Nil

Figure 6. Color Coding for Resistors. From. *http://www.circuitstoday.com/wp-content/uploads/2014/03/6-Band_Resistor_Color_Code-782x1024.jpg*

RESISTANCE

Ohm's law:

$$\vec{E} = \rho\vec{J} \quad \Rightarrow \quad \vec{\mathbf{J}} = \sigma\vec{\mathbf{E}}$$

ρ is constant and independent of the electric field

V and total current I is easier to measure than E and J

RESISTANCE

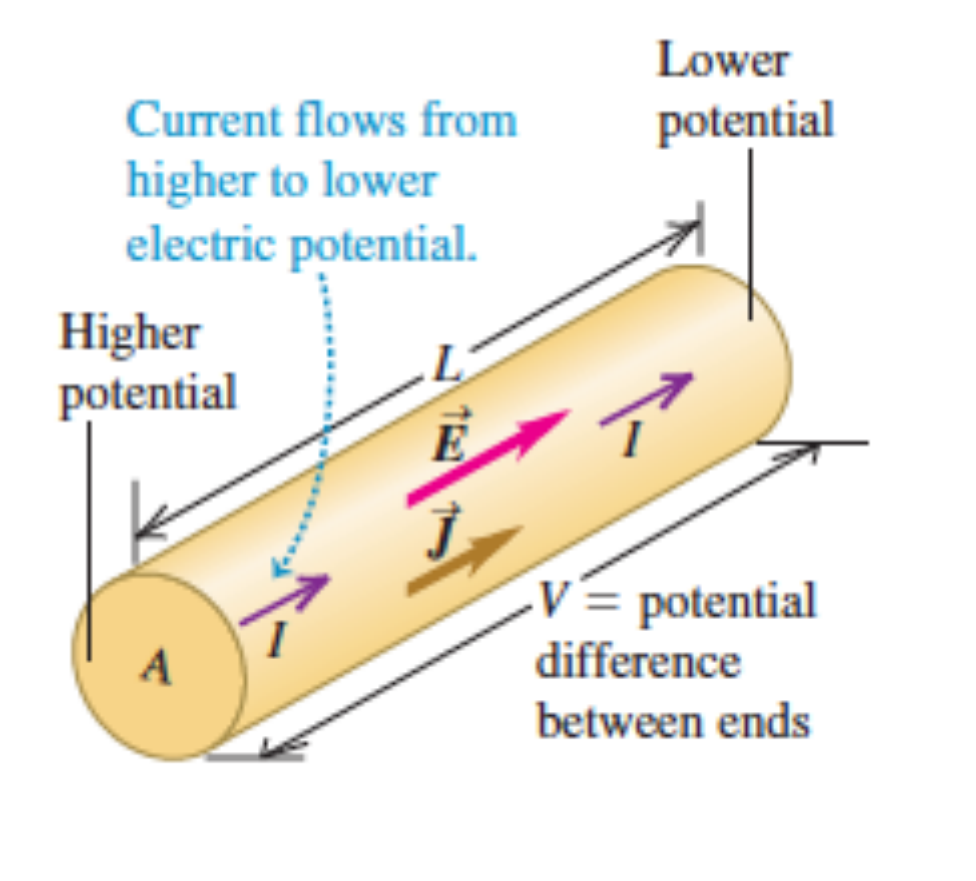


Figure 7. Current density is uniform in any cross section. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 825) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

USEFUL EQUATIONS:

$$V = EL$$

$$E = \rho J$$

$$\mathrm{J} = \frac{\mathrm{I}}{\mathrm{A}}$$

$$V = \frac{\rho L}{A} I$$

RESISTANCE

- The ratio of V per current I is called the resistance of the conductor:

$$R = \frac{V}{I} \quad \Rightarrow \quad V = IR$$

- The relationship of the resistivity to the potential is:

$$R = \frac{\rho L}{A}$$

Depends on the dimensions and material

RESISTANCE

- Resistance also varies with temperature

$$R(T) = R_o\left[1 + \alpha(T - T_o)\right]$$

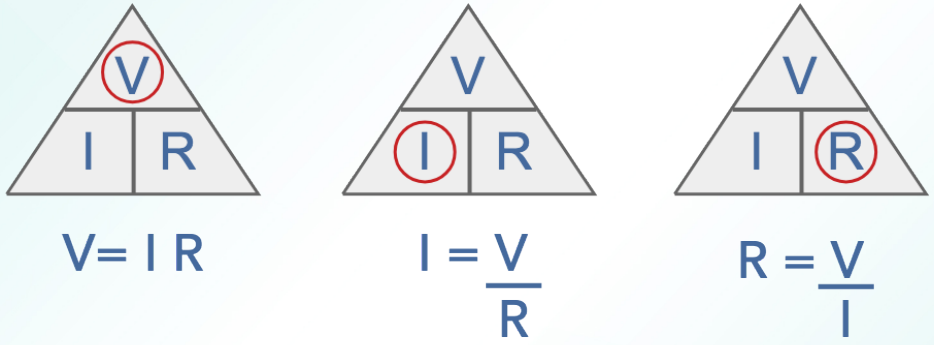
- R_o is the resistivity at a reference temperature T_o (0^oC or 20^oC)

α is the temperature coefficient of resistance

OHM'S LAW

The basic law of current flow is Ohm's law. It is named after its discoverer, the German physicist Georg Ohm.

The ratio of the potential difference to the current is called the resistance of the conductor, and is measured in ohms, symbol Ω.



> **The electric current flowing through a metallic conductor is directly proportional to the potential difference between its ends, as long as the temperature stays the same.**

OHM'S LAW

The relationship between voltage, current and resistance can be simply described by what we call Ohm's Law.

Ohm's Law states that: "*the current flowing in a circuit is directly proportional to the voltage applied and inversely proportional to the resistance, given that the temperature is held constant*"

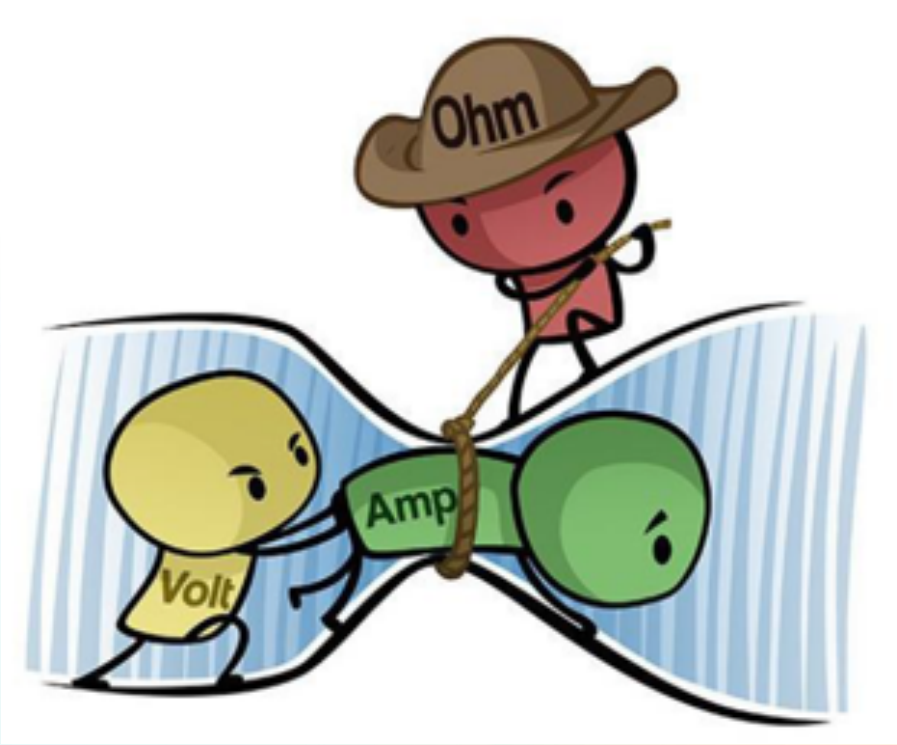


Figure 8. Ohm's Law. From. *https://cdn.instructables.com/FV7/BX1U/JN8YDEFY/FV7BX1UJN8YDEFY.LARGE.jpg*

Example 1 – Ohm's Law

A light bulb has a resistance of 250Ω. Find the current in it when it is placed in a110V circuit.

Given: **V = 110V** **R = 250Ω**

Required: **I = ?**

$$R = \frac{V}{I}$$

$$I = \frac{V}{R} = \frac{110V}{250\Omega}$$

$$I = 0.44\ A$$

Example 2 – Ohm's Law

An electric toaster is connected to a power source of 220V. The current through it is 0.5A. What is its resistance?

Given: **V = 220V** **I = 0.50A**

Required: **R = ?**

$$R = \frac{V}{I}$$

$$R = \frac{V}{I} = \frac{220V}{0.50A}$$

$$R = 440\ \Omega$$

Example 3 – Ohm's Law

What potential difference must be applied across a 1200Ω resistor in order that the resulting current be 30 milliamperes?

Given: **R = 1200Ω** **I = 30 mA**

Required: **V = ?**

$$R = \frac{V}{I}$$

$$V = IR$$

$$V = (3x10^{-3}A)(1200\Omega)$$

$$V = 36V$$

REFERENCES

Serway, R. A., & Jewett, J. W. (2019). Physics for Scientists and Engineers with Modern Physics Tenth Edition. Australia: Cengage Learning.

Young, H. D., & Freedman, R. A. (2016). Sear's & Zemansky's University Physics With Modern Physics. England: Pearson.

http://s3-ap-southeast-1.amazonaws.com/subscriber.images/physics/2016/10/27125734/Current-Density.png

https://tse1.mm.bing.net/th?id=OIP.sFSTlowlBvCRq9CZ9O0LIQHaD5&pid=Api&P=0&w=344&h=182

REFERENCES

http://2.bp.blogspot.com/-oV6vCLqwnc8/TpFltQmBw9I/AAAAAAAAAEY/qHdnVQwSfI4/s1600/Current+Effects.JPG

http://www.circuitstoday.com/wp-content/uploads/2014/03/6-Band_Resistor_Color_Code-782x1024.jpg

https://cdn.instructables.com/FV7/BX1U/JN8YDEFY/FV7BX1UJN8YDEFY.LARGE.jpg

College Physics 2

Power, Energy, EMF and Internal Resistance

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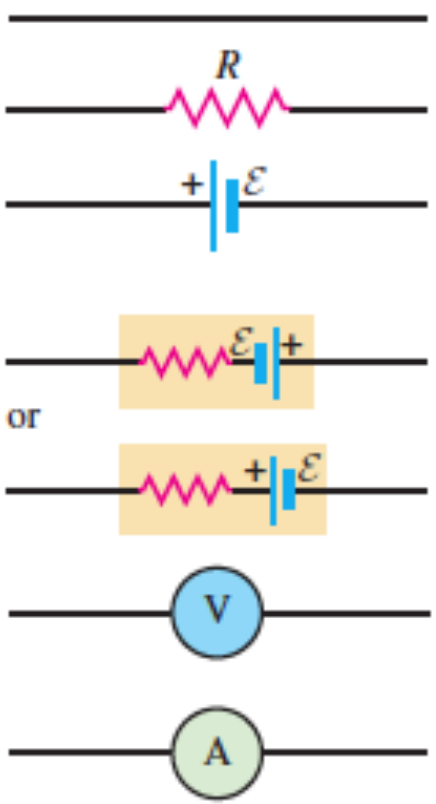
OBJECTIVE

- Solve the energy and power in an electric circuit.
- Understand the effect of emf and internal resistance in a circuit

Symbols for Circuit Diagrams



Conductor with negligible resistance

Resistor

Source of emf (longer vertical line always represents the positive terminal, usually the terminal with higher potential)

Source of emf with internal resistance *r* (*r* can be placed on either side)

Voltmeter (measures potential difference between its terminals)

Ammeter (measures current through it)

Figure 9. Symbols for Circuit Diagrams .From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 831) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

POWER

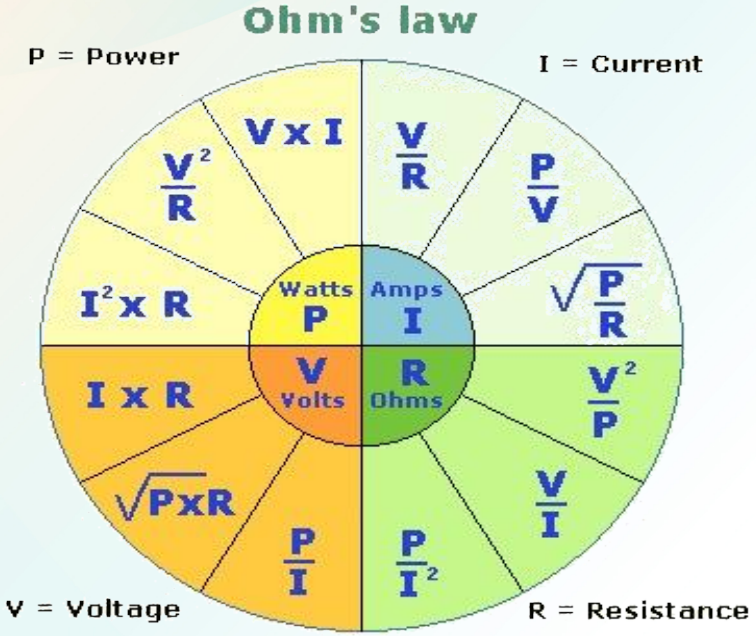


Figure 10. Ohms' Law Wheel. From. *http://www.reuk.co.uk/OtherImages/ohms-law.gif*

Electrical power is the rate at which electrical energy is being used.

$$P = VI$$

P = Power, W or J/s V = Potential, V

I = Current, A

Sample Problem

1. Find the power consumed by a 120 Ω light bulb when the current through it is 0.9 A.

Given: $R = 120\ \Omega$ $I = 0.9\ A$

Required: P

Solution:

$$P = I^2R$$

$$P = (0.9\ A)^2\ (120\ \Omega)$$

$$P = 97.2\ \text{Watts}$$

Sample Problem

2. The power dissipated by an electric appliance is 30W. What is the current flowing through it if it is connected to a power source of 110V?

Given: R = 30 W, V =110V
Required: I
Solution:

$P = VI$

$I = P / V$

$I = (30W/110V)$ $I = 0.27\ A$

Power Output of a Source

- A source of electromotive force (emf) provides energy

$$P = IV_{ab}$$

- For an emf source that can be described by an **internal resistance** *r*

$$V_{ab} = \mathcal{E} - Ir$$

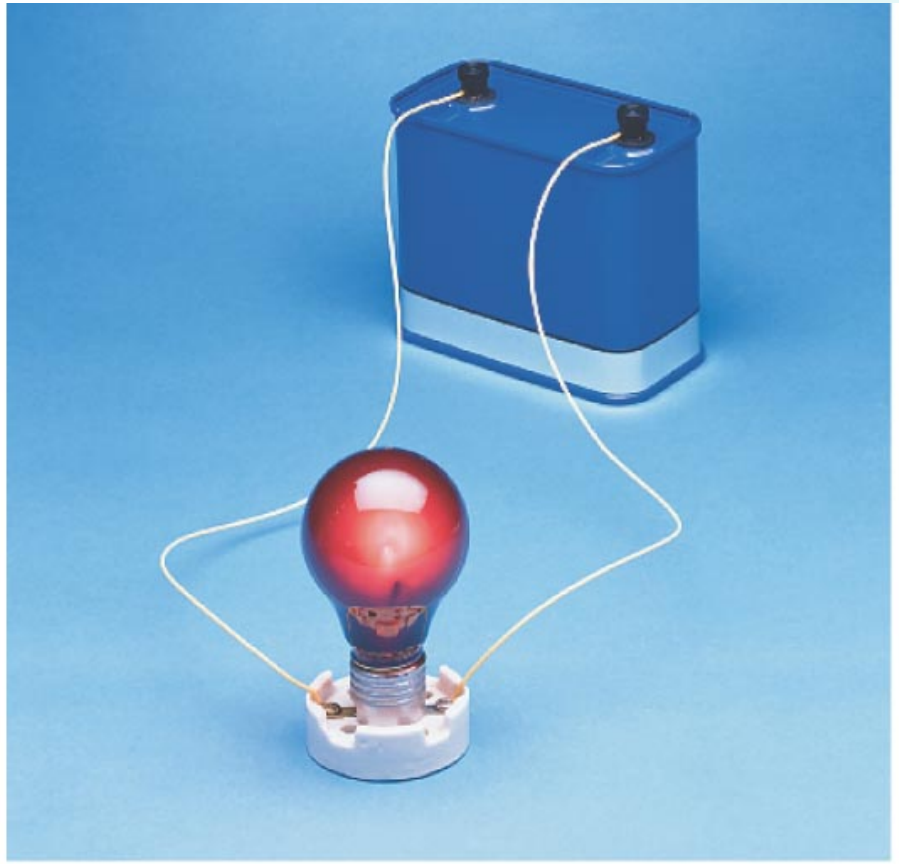


Figure 11. Real Source of emf and a bulb.. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 831) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

Power Input to a Source

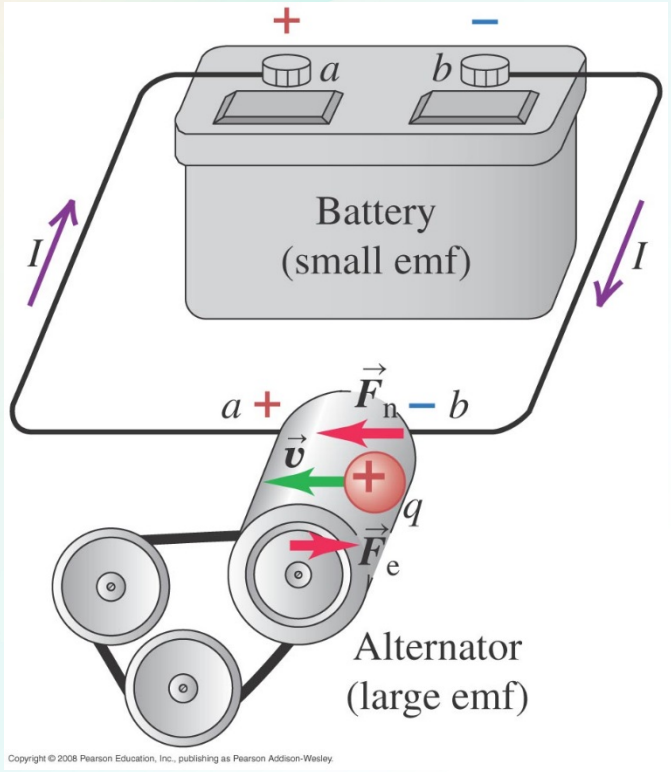


Figure 11. Power input to a source.. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 835) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

Rate of conversion from non-electrical to electrical energy

$$P = I(\mathcal{E} + Ir) = \mathcal{E}I + I^2 r$$

Net power **input to the** source

Rate at which electric energy is dissipated in the internal resistance

- If the emf source is connected to a larger emf source (battery charging)

Sample Problem

3. Given the figure, what are the rates of energy conversion and dissipation and the net power output of the battery?

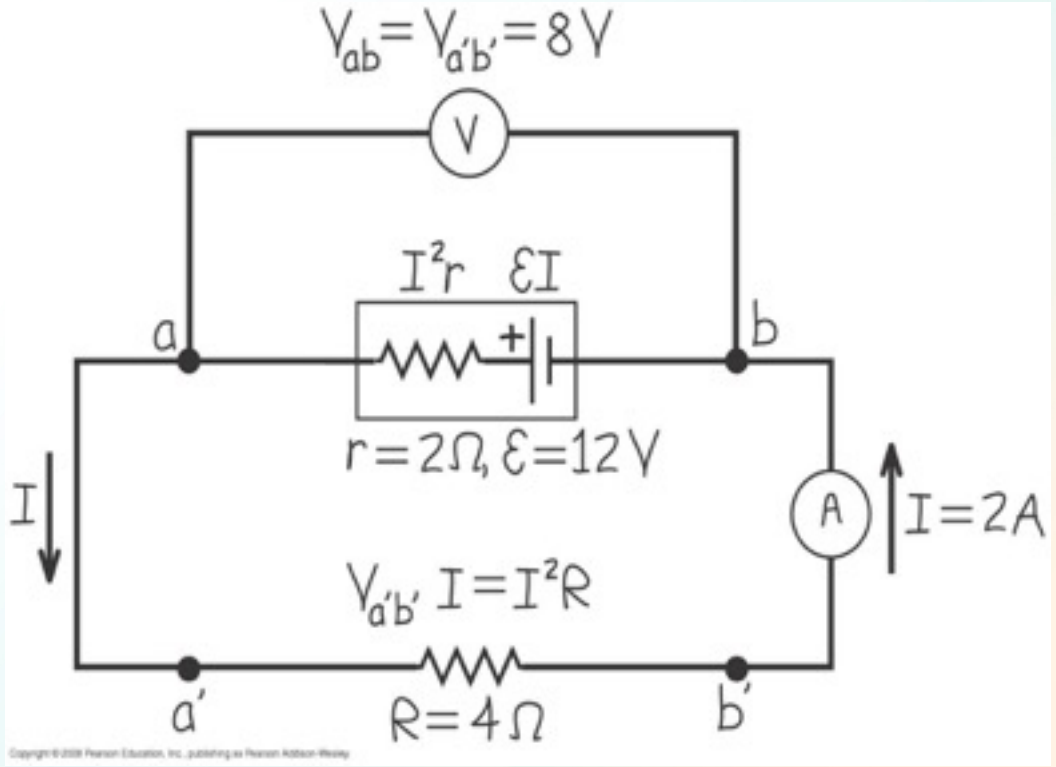


Figure 12. Power input to a source.. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 836) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

Sample Problem

Solution:

a) Energy conversion:

$$\mathcal{E}I = (12V)(2A) \quad = 24\ W$$

b) Energy dissipation:

$$I^2 r = (2A)^2(2\Omega) = 8W$$

c) Net power output:

$$\mathcal{E}I - I^2 r = (24W) - (8W) = 16W$$

REFERENCES

Serway, R. A., & Jewett, J. W. (2019). Physics for Scientists and Engineers with Modern Physics Tenth Edition. Australia: Cengage Learning.

Young, H. D., & Freedman, R. A. (2016). Sear's & Zemansky's University Physics With Modern Physics. England: Pearson.

http://www.reuk.co.uk/OtherImages/ohms-law.gif

College Physics 2

Series Parallel Resistors

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OBJECTIVES

- Give the characteristics of series and parallel resistive circuits
- Compute the equivalent resistance, current, and emf in series or parallel or series-parallel connection

ELECTRIC CIRCUIT

Electric circuit is a path of an electric current.

The term is usually taken to mean a continuous path composed of conductors and conducting devices and including a source of electromotive force (emf) that drives the current around the circuit. A circuit of this type is termed a closed circuit, and a circuit in which the current path is not continuous is called an open circuit.

A short circuit is a closed circuit in which a direct connection is made, with no appreciable resistance, inductance, or capacitance, between the terminals of the source of electromotive force.

Resistance in Series

Electric circuits with only a single path for electricity to flow along are called series circuits.

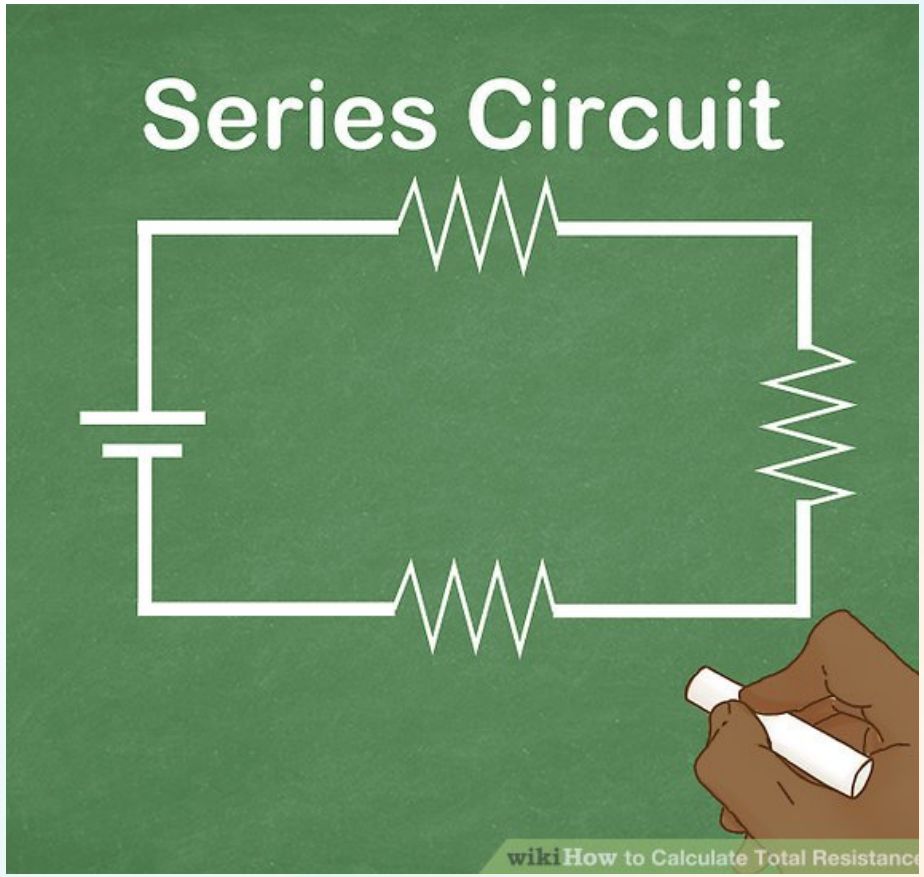


Figure13. Resistors in Series. From. *https://www.wikihow.com/images/thumb/8/8e/Calculate-Total-Resistance-in-Circuits-Step-1-Version-2.jpg/aid487580-v4-728px-Calculate-Total-Resistance-in-Circuits-Step-1-Version-2.jpg*

Resistance in Series

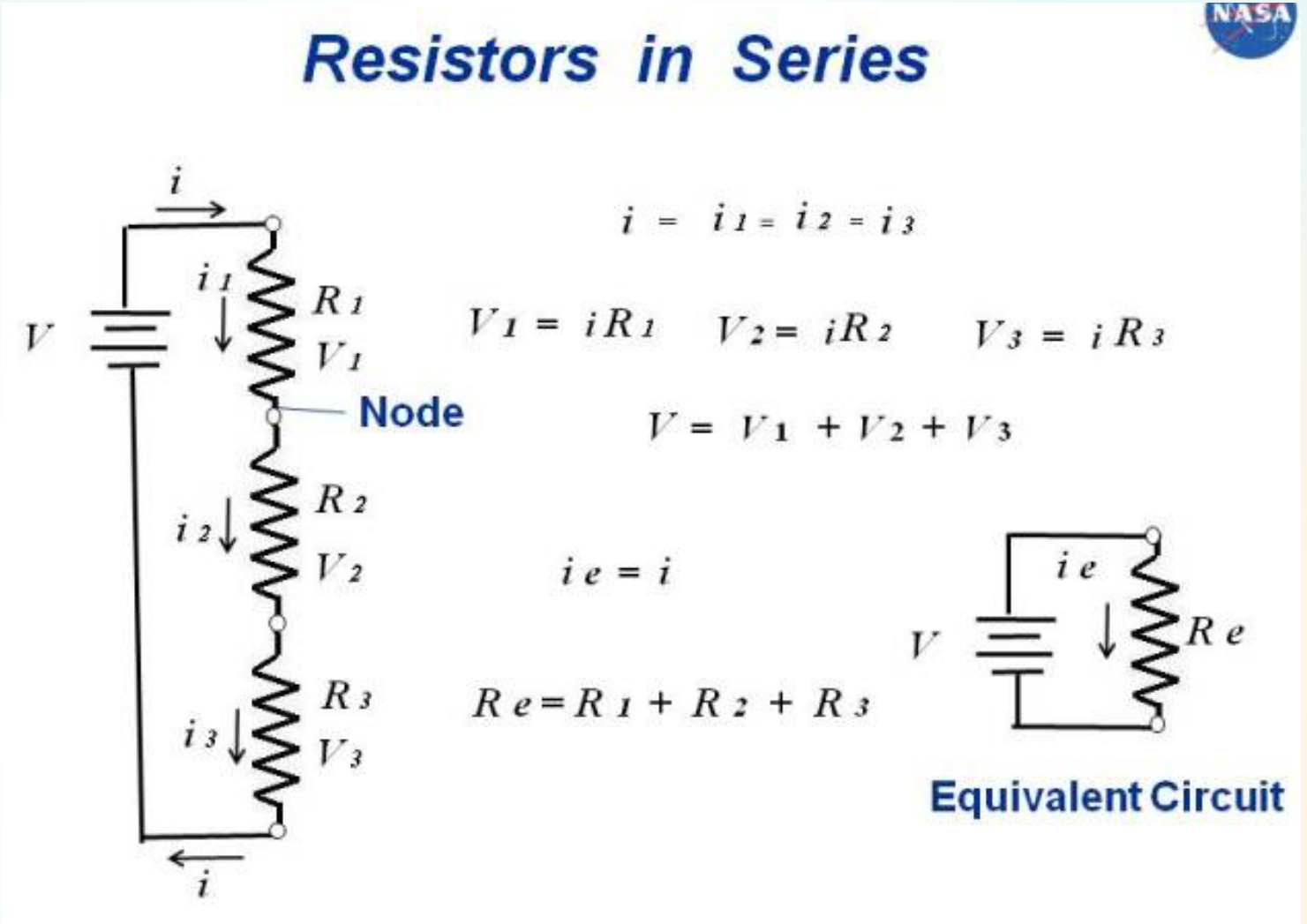


Figure 14. Resistance in Series Equations. From. *https://www.grc.nasa.gov/www/*k-12/airplane/Images/ohmrser.jpg

Resistance in Parallel

Electric circuits with multiple pathways for electricity to flow along are called parallel circuits.

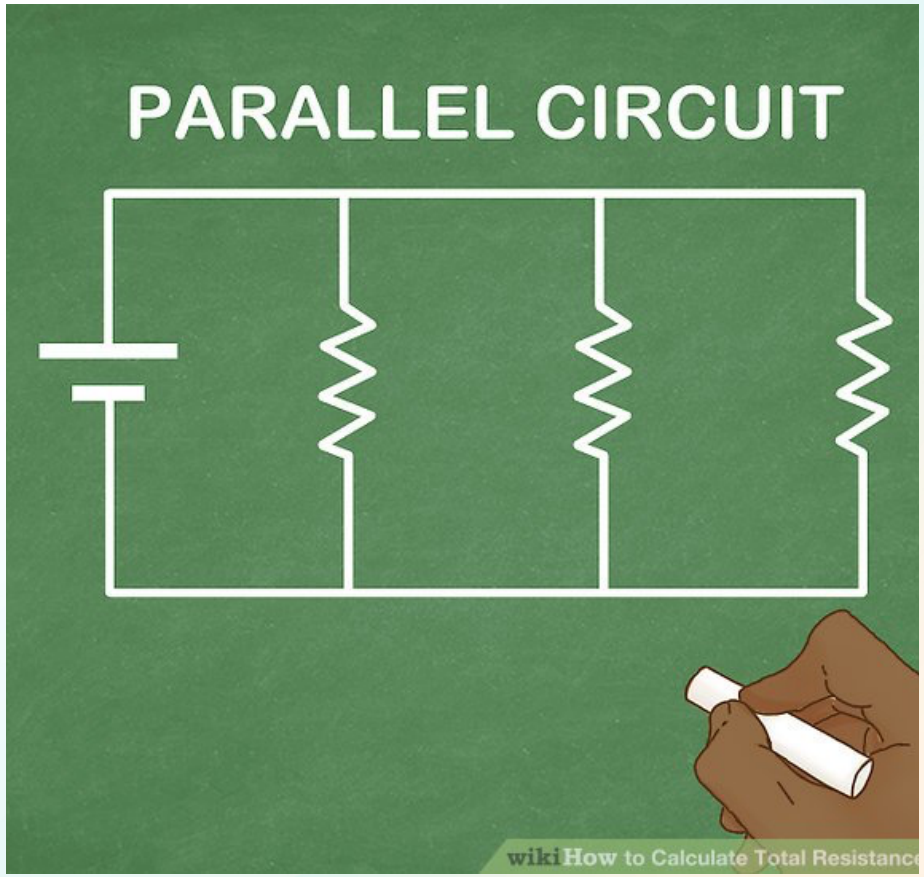


*Figure 15.*Resistors in Parallel. From. *https://www.wikihow.com/images/thumb/0/0a/Calculate-Total-Resistance-in-Circuits-Step-5-Version-2.jpg/aid487580-v4-728px-Calculate-Total-Resistance-in-Circuits-Step-5-Version-2.jpg*

Resistance in Parallel

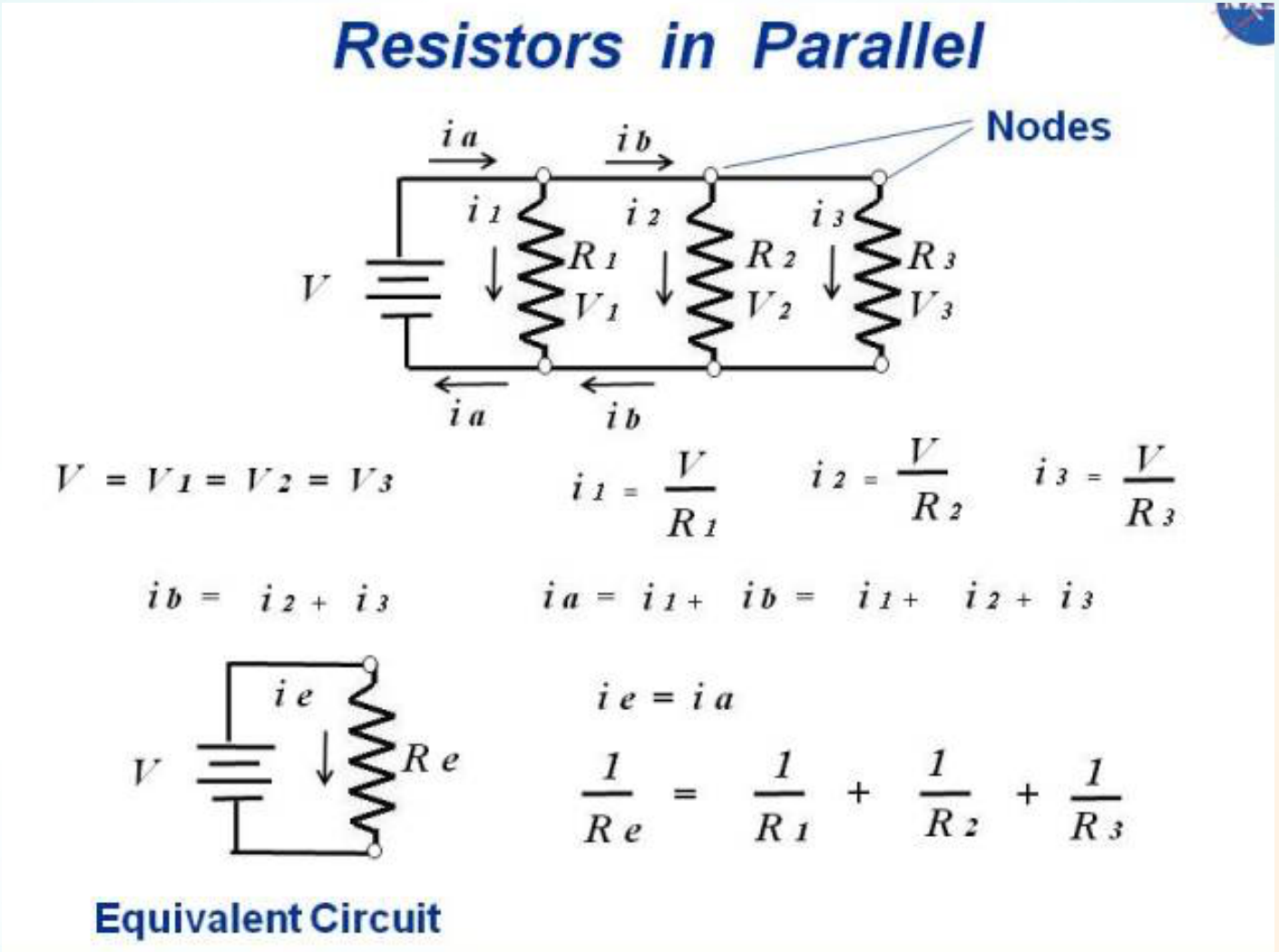
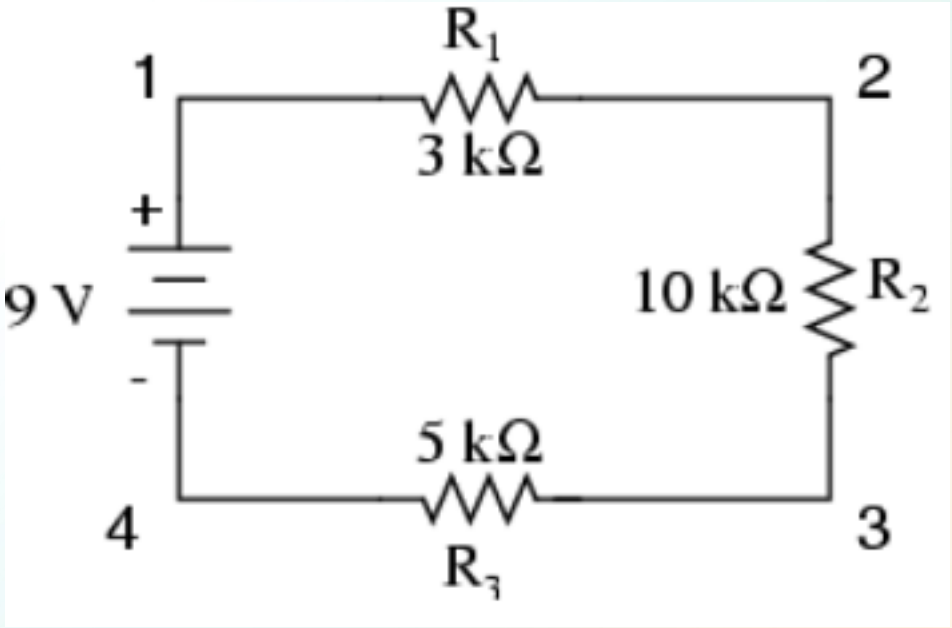


Figure 16. Resistance in Parallel Equations. From. *http://www.grc.nasa.gov/WWW/K-12/airplane/Images/ohmrpar.jpg*

Sample Problems

1. Find the total resistance of the given circuit.



ANSWER: 18 kΩ

Sample Problems

2. Three resistors 2Ω, 3Ω, and 5Ω are connected in (a)series (b) parallel.
Find the equivalent resistance, the current in each resistor and the potential difference across each resistor if the network is connected to a power source of 50V.

2. Solution

a) RESISTORS IN SERIES

R_T

I_1, I_2, I_3

V_1, V_2, V_3=?

For R_T:

$$R_T = R_1 + R_2 + R_3$$
$$R_T = (2\Omega + 3\Omega + 5\Omega)$$
$$R_T = 10\Omega$$

For I_T:

$$I_T = \frac{V_T}{R_T} = \frac{50V}{10\Omega} = 5A$$

Since: $I_T = I_1 = I_2 = I_3 = 5A$

$$V_1 = I_1 R_1 = (5A)(2\Omega) = 10V$$

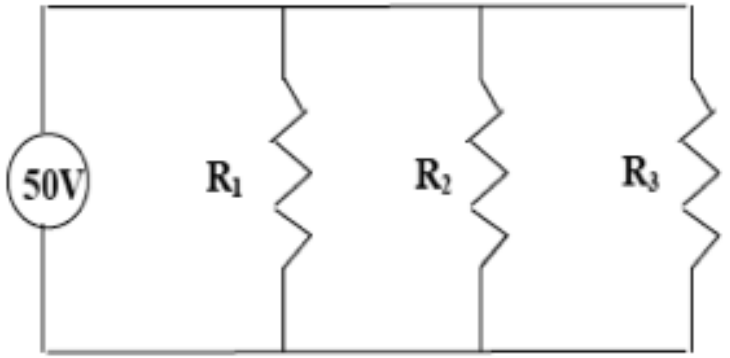
$$V_2 = I_2 R_2 = (5A)(3\Omega) = 15V$$

$$V_3 = I_3 R_3 = (5A)(5\Omega) = 25V$$

2. Solution

b) RESISTORS IN PARALLEL

R_T,
I_1, I_2, I_3,
V_1, V_2, V_3=?



$$V_T = V_1 = V_2 = V_3 = 50V$$

$$I_1 = \frac{V_1}{R_1} = \frac{50V}{2\Omega} = 25A$$

For R_T:

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$I_2 = \frac{V_2}{R_2} = \frac{50V}{3\Omega} = 16.67A$$

$$\frac{1}{R_T} = \frac{1}{2\Omega} + \frac{1}{3\Omega} + \frac{1}{5\Omega}$$

$$I_3 = \frac{V_3}{R_3} = \frac{50V}{5\Omega} = 10A$$

RT = 0.97Ω

REFERENCES

Serway, R. A., & Jewett, J. W. (2019). Physics for Scientists and Engineers with Modern Physics Tenth Edition. Australia: Cengage Learning.

Young, H. D., & Freedman, R. A. (2016). Sear's & Zemansky's University Physics With Modern Physics. England: Pearson.

https://i.ytimg.com/vi/ppZMG7y4kg8/maxresdefault.jpg

https://www.grc.nasa.gov/www/k-12/airplane/Images/ohmrser.jpg

http://www.grc.nasa.gov/WWW/K-12/airplane/Images/ohmrpar.jpg

College Physics 2

Kirchhoff's Law

MPS Department | FEU Institute of Technology

OBJECTIVES

- Explain Kirchhoff's loop and current rules.
- Use Kirchhoff's rules in analyzing and solving resistive circuits.
- Analyze circuits that include both a resistor and a capacitor
- Solve RC circuits problems

Kirchhoff's Laws

Kirchhoff's Voltage Law(KVL): *The algebraic sum of the potential differences in any closed loop must equal to zero.*

$$\sum V = 0$$

Kirchhoff's Current Law (KCL): *The algebraic sum of the currents into any junction is equal to zero.* Or in other words, *current ENTERING into a junction is equal to the CURRENT leaving the junction.*

$$\sum I = 0 \; ; \; I_{entering} = I_{leaving}$$

Kirchhoff's Laws

Two networks that cannot be reduced to simple series-parallel combinations of resistors.

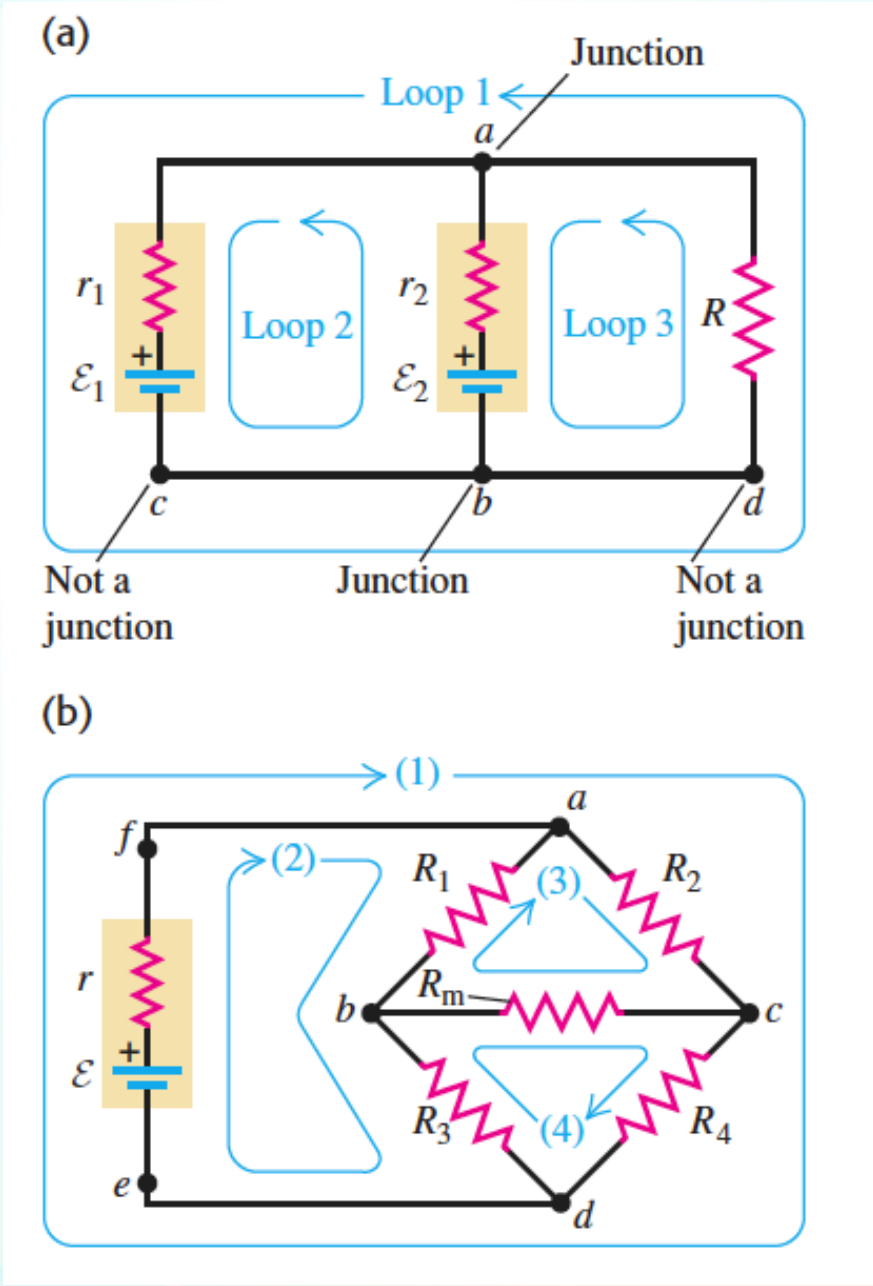
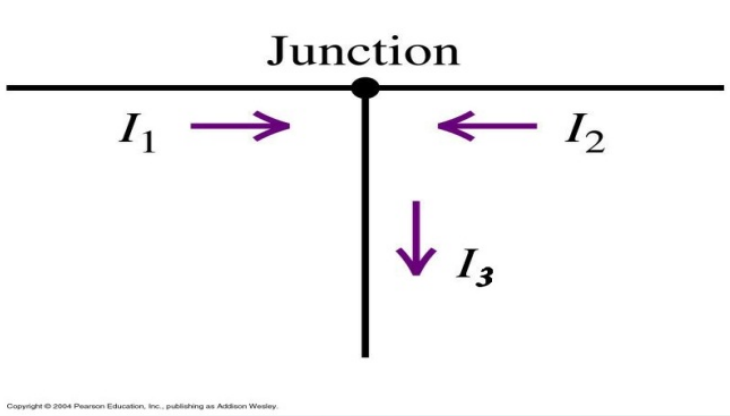


Figure 17 a) & b) Two networks that can not be reduced. From. . Real Source of emf and a bulb.. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 855) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

Kirchhoff's Laws

A *junction*, also called a *node* or *branch point*, is a point where three or more conductors meet

A *loop* is any closed conducting path



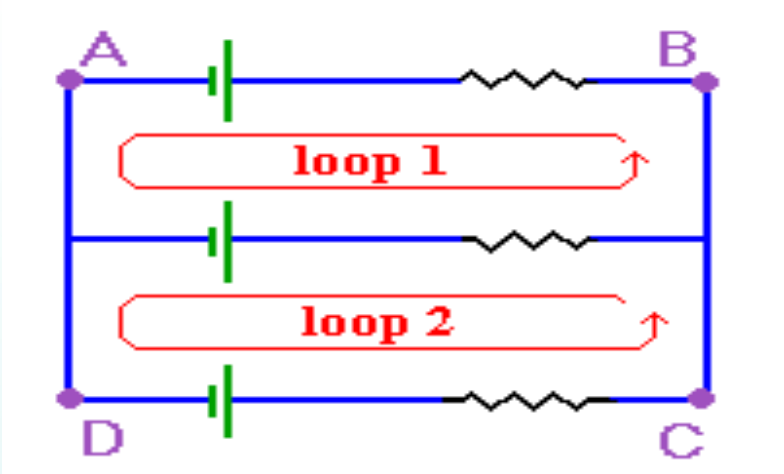


Figure 18. Kirchhoff's junction /loop rule. . *From.* . Real Source of emf and a bulb.. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 856) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

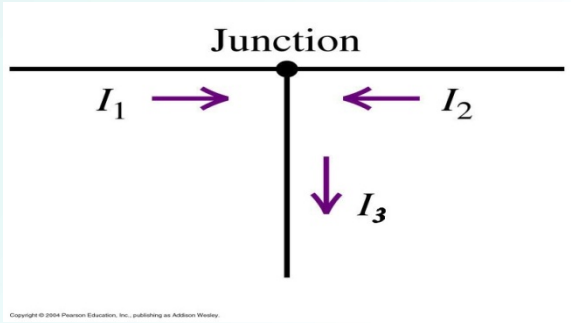
Kirchhoff's Rules are basically two statements

1. The algebraic sum of the currents into any junction is zero $\sum I = 0$

A sign convention:

A current heading *towards* a junction, is considered to be *positive*,

A current heading *away* from a junction, is considered to be *negative*



$$I_1 + I_2 - I_3 = 0$$

Be aware that all the junction equations for a circuit may not be independent of each other

Kirchhoff's Rules are basically two statements

2. The algebraic sum of the potential differences in any loop including those associated with emfs and those of resistive elements must equal zero

$$\sum V = 0$$

Procedures to apply this rule:

Pick a direction for the current in each branch

If you picked the wrong direction, the current would come out negative

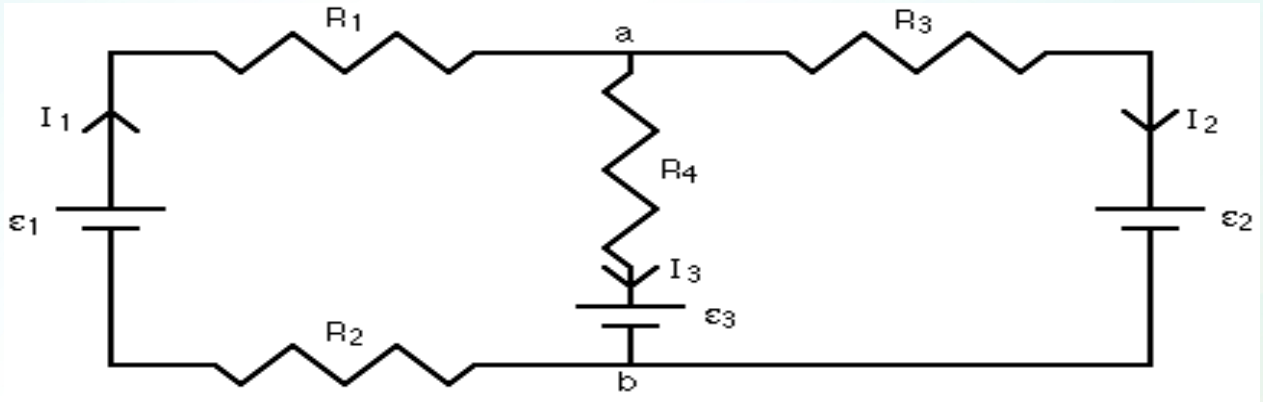
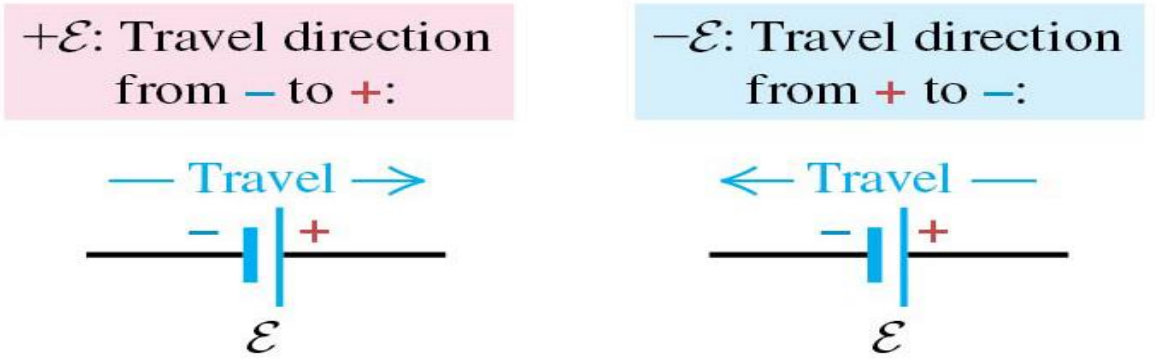


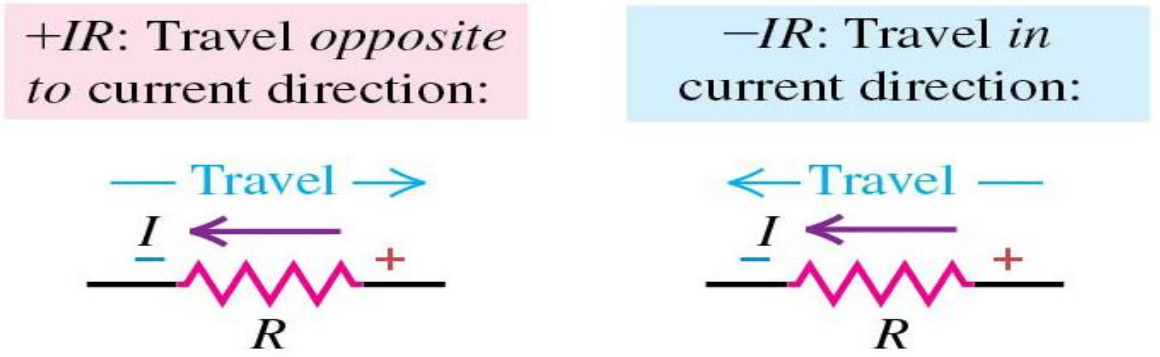
Figure 19. Multiloop Circuits. From. *http://buphy.bu.edu/~duffy/PY106/10b.GIF*

Sign conventions for the loop rule

- Use these sign conventions when you apply Kirchhoff's loop rule.



- In each part of the figure, "Travel" is the direction that we imagine going around the loop, which is not necessarily the direction of the current.



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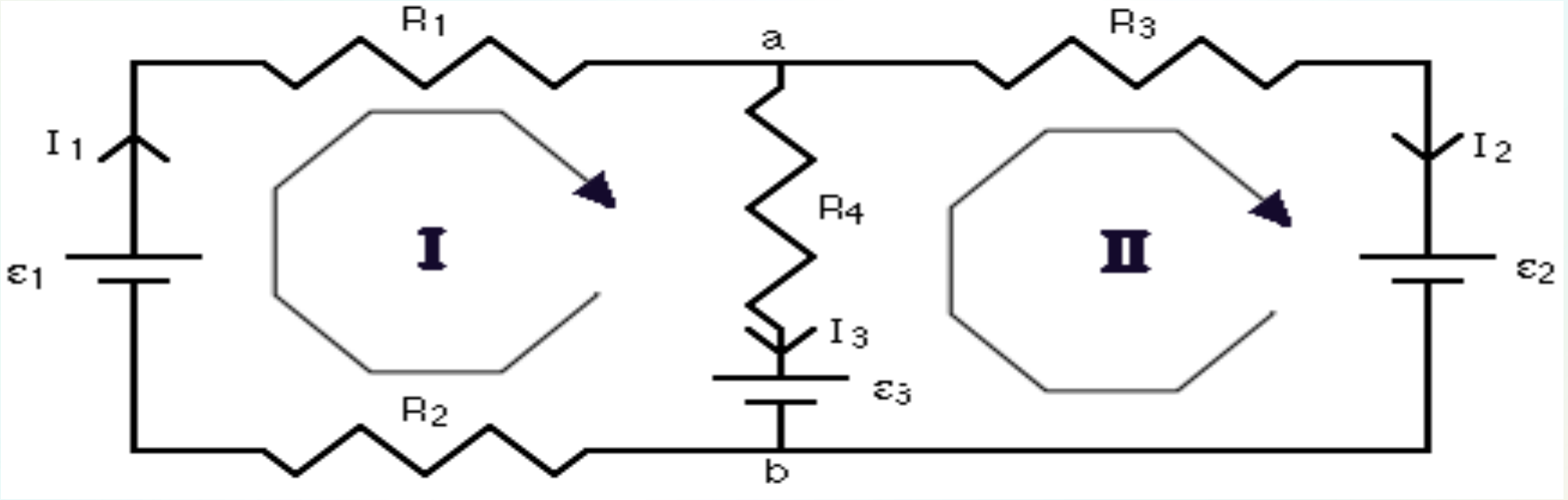
Kirchoff's Rules

Pick a direction for traversing a loop – this direction must be the same for all loops



Note that there is a third loop along the outside branches

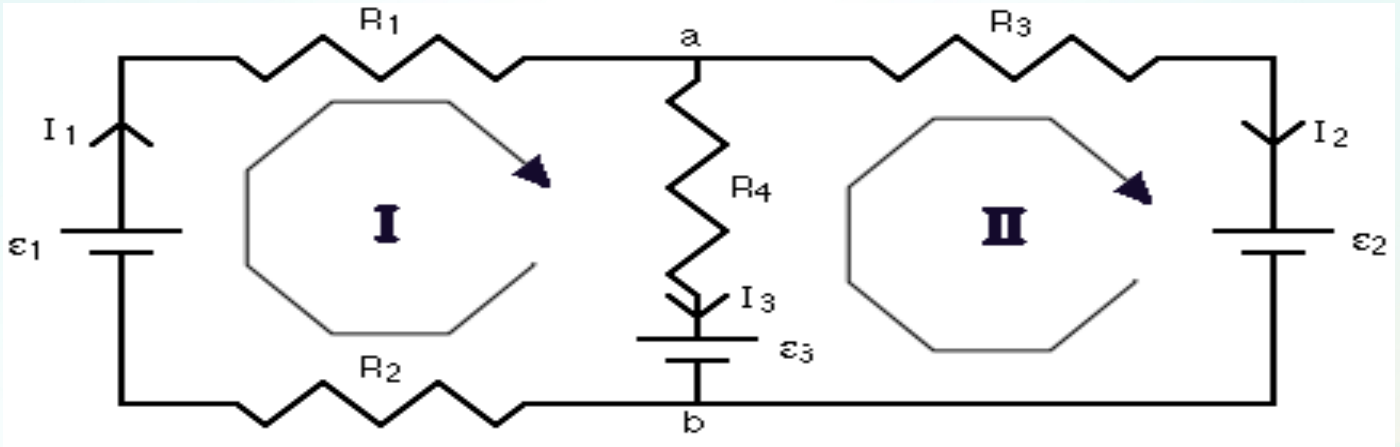
As with the junction equations not all the loop equations will be independent of each other.

Kirchoff's Rules

Starting at any point on the loop add the emfs and *IR* terms

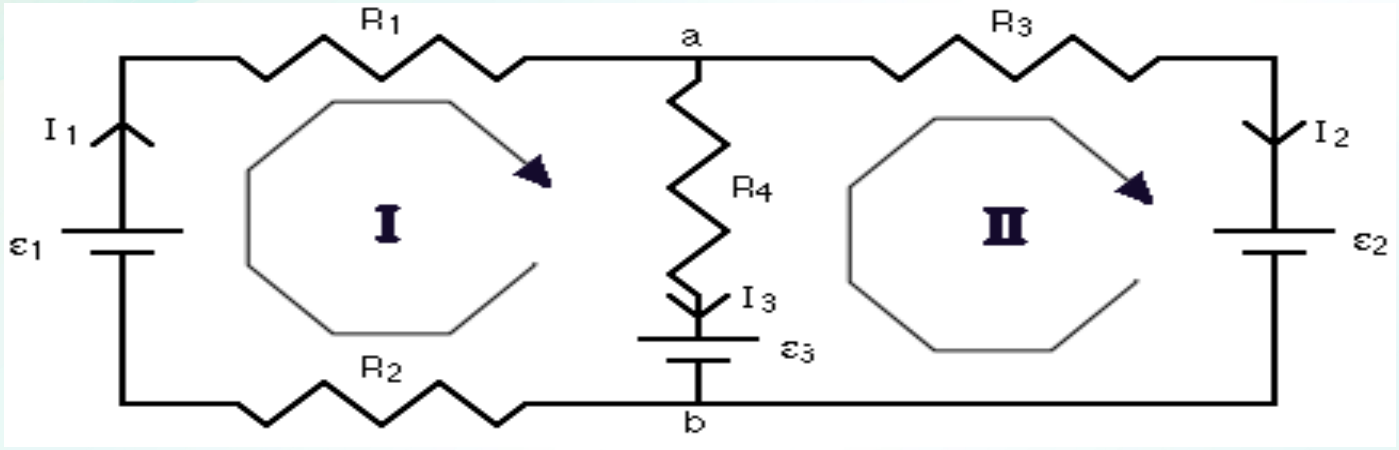
An IR term is negative if we traverse it in the same sense as the current that is going through it, otherwise it is positive

An emf is considered to be positive if we go in the direction - to +, otherwise it is negative



Need to have as many independent equations as there are unknowns

Kirchoff's Rules



For loop I we have

$$-I_1R_1 - I_1R_2 - I_3R_4 + \varepsilon_1 - \varepsilon_3 = 0$$

For loop II we have

$$-I_2R_3 + I_3R_4 - \varepsilon_2 + \varepsilon_3 = 0$$

Junction equation at *a* gives us

$$I_1 - I_2 - I_3 = 0$$

We now have three equations for the three unknown currents

Sample Problem

1. A single loop circuit contains two resistors and two batteries as shown in the figure.
Find the current in the circuit.

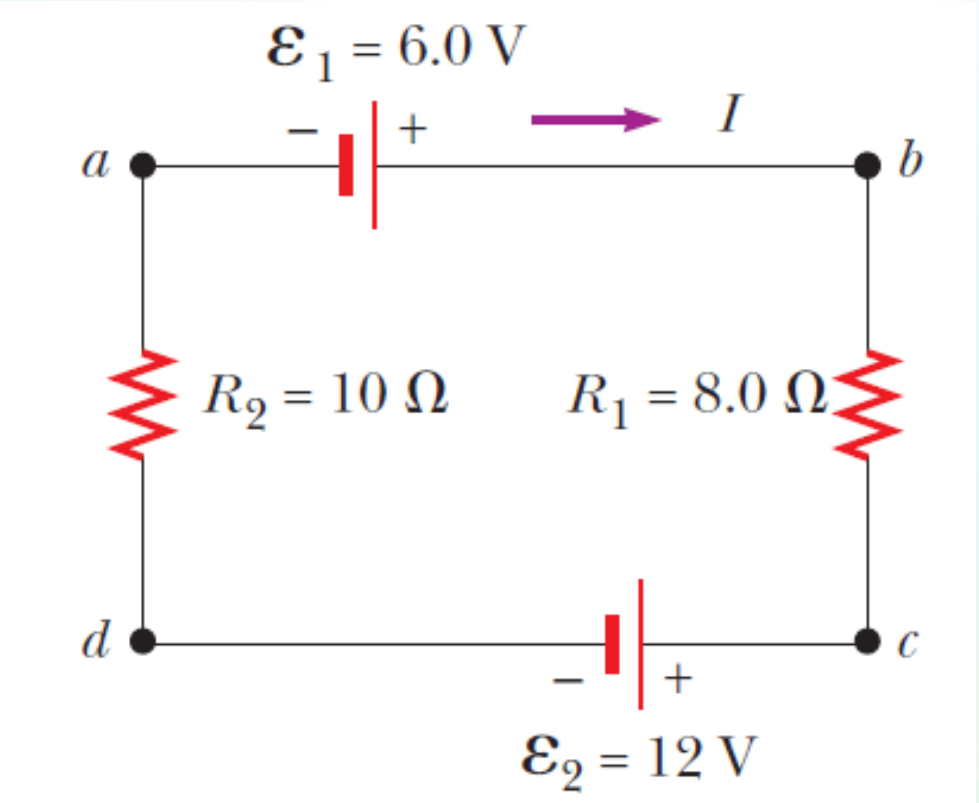
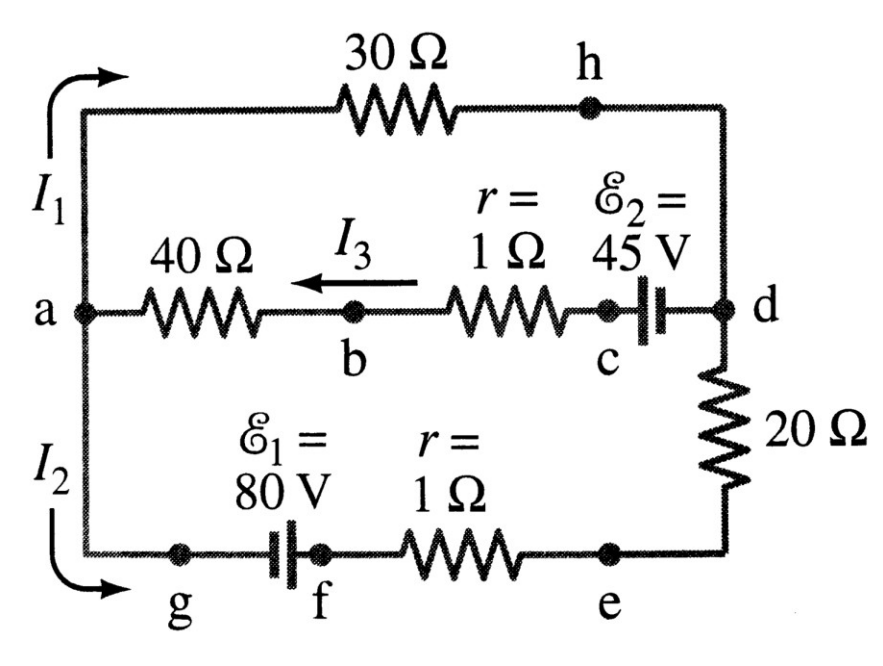


Figure 19. Sample Problem. . From. *Physics for Scientists and Engineers with Modern Physics*(pp 845) by Serway, R. A., & Jewett, J. W. (2019). Australia: Cengage Learning.

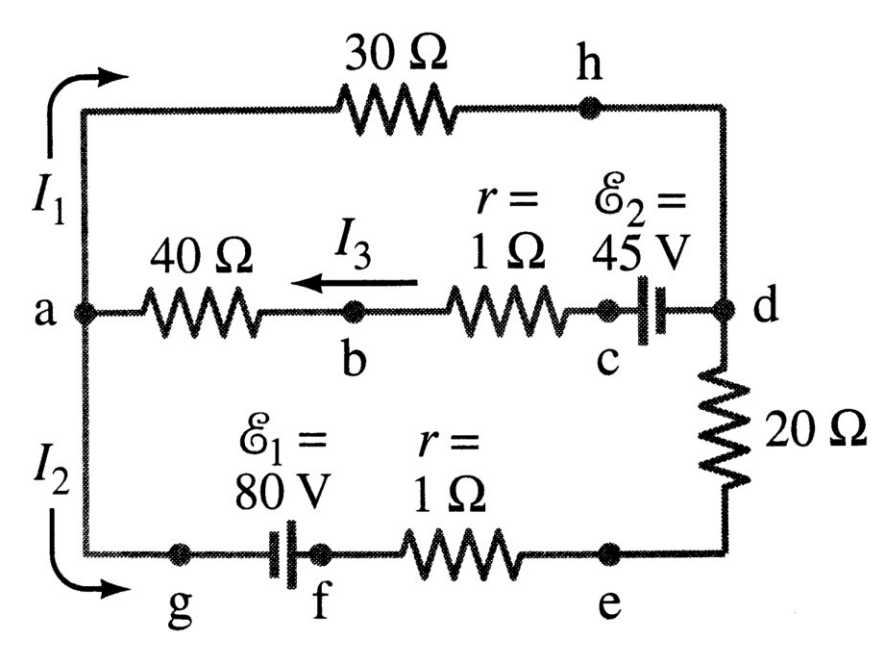
ANS: -0.33 A

MULTILOOP CIRCUITS



Calculate the currents I_1, I_2 and I_3 in the 3 branches of the circuit.

MULTILOOP CIRCUITS

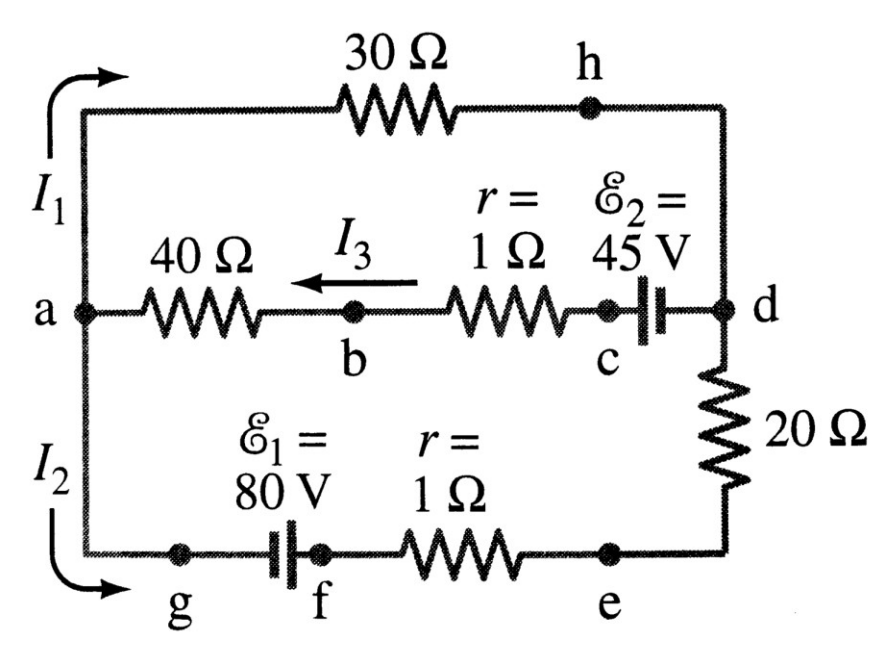


Calculate the currents I_1, I_2 and I_3 in the 3 branches of the circuit.

1. **Label the currents**
    1. Use I_1, I_2, I_3 for the current in 3 separate branches.
    2. The directions of I_2 and I_3 are chosen as such since current tends to move away from the positive terminal of a battery.
    3. Direction of I_1 is not obvious so an arbitrary direction is assumed. If the computed I_1 is negative, then the direction should be the opposite.
2. **Identify the unknowns (I_1, I_2 and I_3)**

There are 3 unknowns so 3 equations are needed to solve.

MULTILOOP CIRCUITS



Calculate the currents I_1, I_2 and I_3 in the 3 branches of the circuit.

3. Junction rule

Junction (a): $\mathbf{I_3 = I_1 + I_2}$ (Eq. 1)

Junction (d): same equation

4. Loop rule

Choose any 2 loops. (3 loops possible)

Loop ahdcba:

$V_{ha} = (-I_1)(30\Omega) = -30I_1$ $V_{dh} = 0$

$V_{cd} = 45$ V $V_{bc} = -I_3(1\Omega) = -I_3$

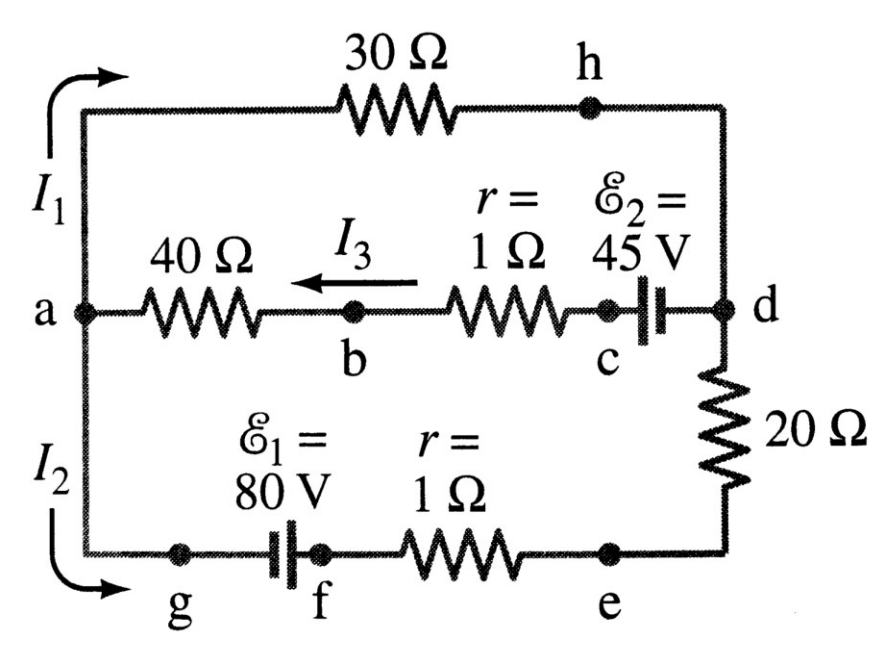
$V_{ab} = -I_3(40\Omega) = -40I_3$

$\mathbf{V_{ha} + V_{dh} + V_{cd} + V_{bc} + V_{ab} = 0}$

$-30I_1 + 0 + 45 + (-I_3) + (-40I_3) = 0$

$\mathbf{-30I_1 + 45 - 41I_3) = 0}$ (Eq. 2)

MULTILOOP CIRCUITS



Calculate the currents I_1, I_2 and I_3 in the 3 branches of the circuit.

4. Loop rule

Loop ahdefga::

$V_{ha}=(-I_1)(30\Omega) = -30I_1$ $V_{dh} = 0$

$V_{ed}= I_2(20\Omega) = 20I_2$ $V_{fe}= I_2(1\Omega) = I_2$

$V_{gf}= -80$ $V_{ga} = 0$

$\mathbf{V_{ha} + V_{dh} + V_{ed} + V_{fe} + V_{gf} + V_{ag} = 0}$

$-30I_1 + 0 + 20I_2 + I_2 + (-80) + 0 = 0$

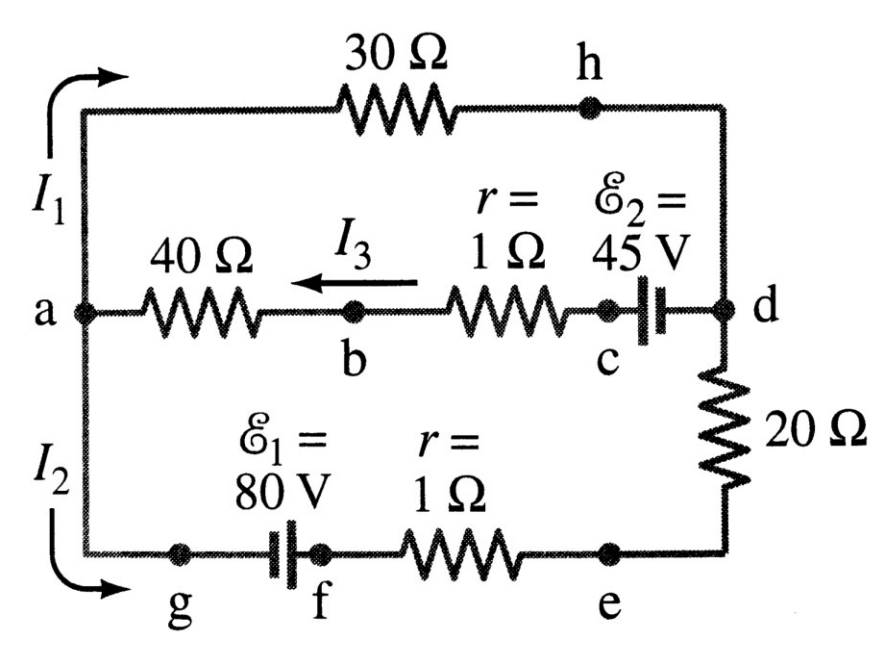
$\mathbf{-30I_1 + 21I_2 - 80) = 0}$ (Eq. 3)

5. Solve the equations

$\mathbf{I_2 = (80+30I_1)/21 = 3.8 + 1.4I_1}$

$\mathbf{I_3 = (45 - 30I_11)/41 = 1.1 - 0.73I_1}$

MULTILOOP CIRCUITS



Calculate the currents I_1, I_2 and I_3 in the 3 branches of the circuit.

5. Solve the equations

$$I_2 = (80+30I_1)/21 = 3.8 + 1.4I_1$$ (Eq. 4)

$$I_3 = (45 - 30I_1 1)/41 = 1.1 - 0.73I_1$$ (Eq. 5)

$$I_1 = I_3 - I_2$$ (Eq. 6)

Substituting Eq. 4) and 5 to Eq. 6,

$I_1 = I_3 - I_2 = 1.1 - 0.73I_1 - 3.8 + 1.4I_1$ **I_1**

$= -0.87A$

The negative sign indicates that the actual direction is opposite to what was initially assumed.

Solving for I_2 and I_3:

$I_2 = 3.8 + 1.4I_1 = 3.8 + 1.4(-0.87)$ **$I_2 = 2.6$ A**

$I_3 = 1.1 - 0.73I_1 = 1.1 - 0.73(-0.87)$ **$I_3 = 1.7$ A**

RC CIRCUITS

Capacitors and resistors are often found together in a circuit. Such RC circuits are used to control a car's windshield wipers and timing of traffic lights, camera flashes, heart pacemakers, and many other electronic devices.

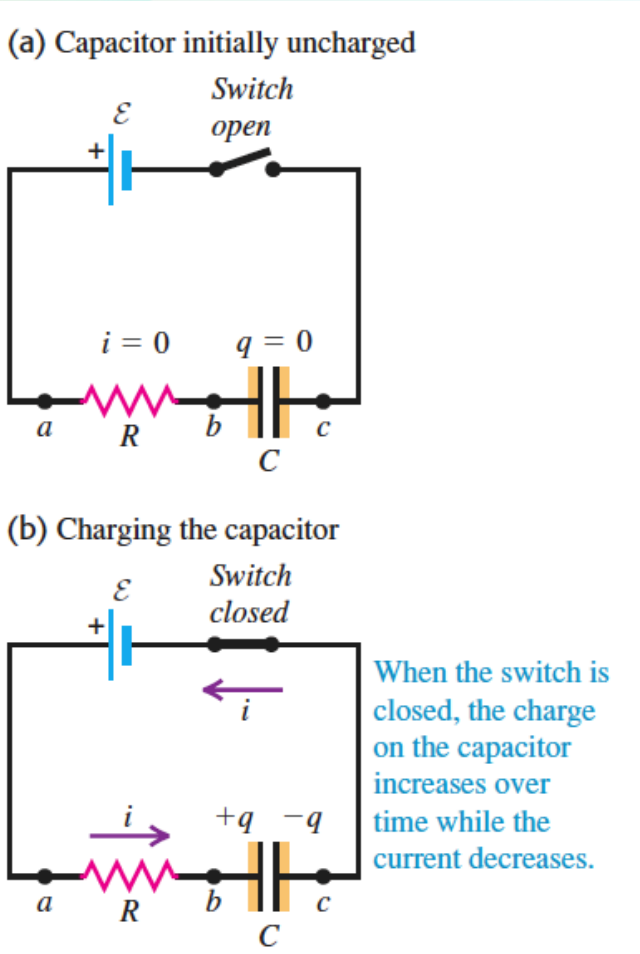


Figure 20 a) Capacitor initially uncharged b) Charging the Capacitor. From. Real Source of emf and a bulb.. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 865) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

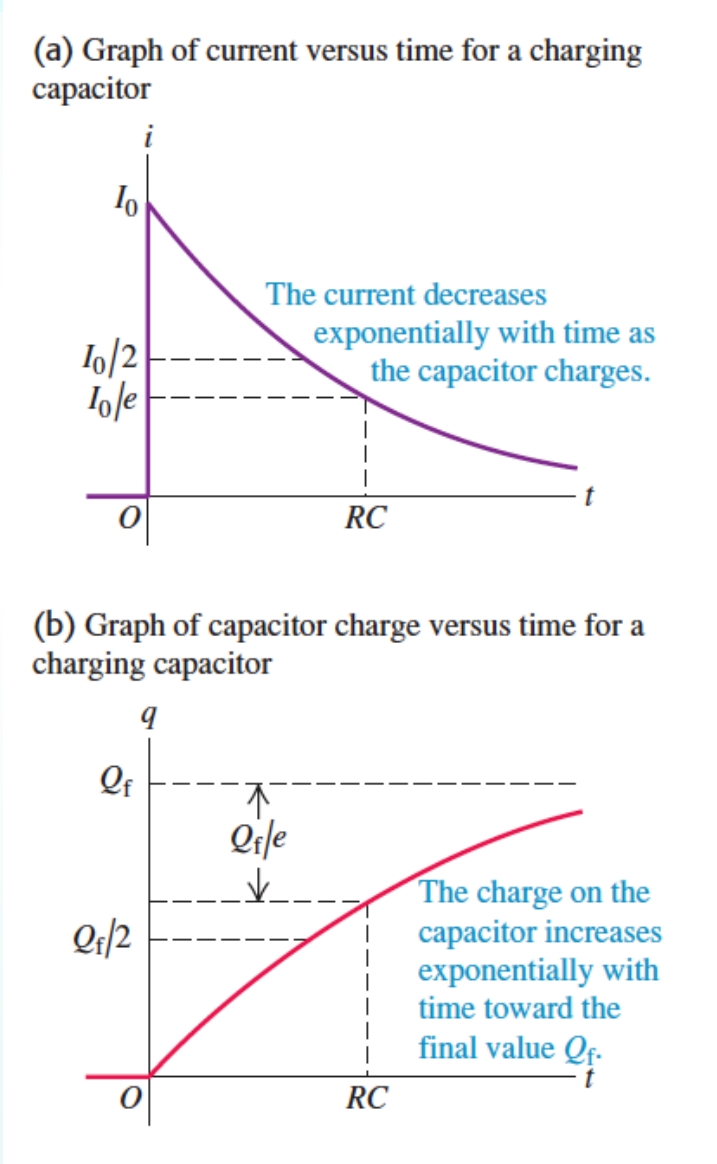


Figure 21 a) Graph of current versus time. b) Graph of capacitor charge versus time.Physics With Modern Physics (pp 865) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson

RC CIRCUITS

CHARGING OF THE CAPACITOR

$$V_C = \frac{Q}{C}$$

$$V_C = \xi(1 - e^{-t/RC})$$

Where V_C = potential difference across capacitor, V
ξ = emf of the battery
Q = charge on the capacitor, C
R= resistance, Ω C
= capacitance, F e
= 2.718
t = time, s
$\tau = RC$= time constant , s (the time it takes to for the voltage to reach 63% of its maximum value)

$$Q = Q_0(1 - e^{-t/RC})$$

$$\tau = RC$$

Where Q_0 = maximum charge, C
R = resistance, Ω
C = capacitance, F
e = 2.718
t = time, s
$\tau = RC$= time constant , s

RC CIRCUITS

For the RC circuit, the voltage on the capacitor decreases with time after the switch S is closed.

The charge on the capacitor follows the same curve since Q αV.

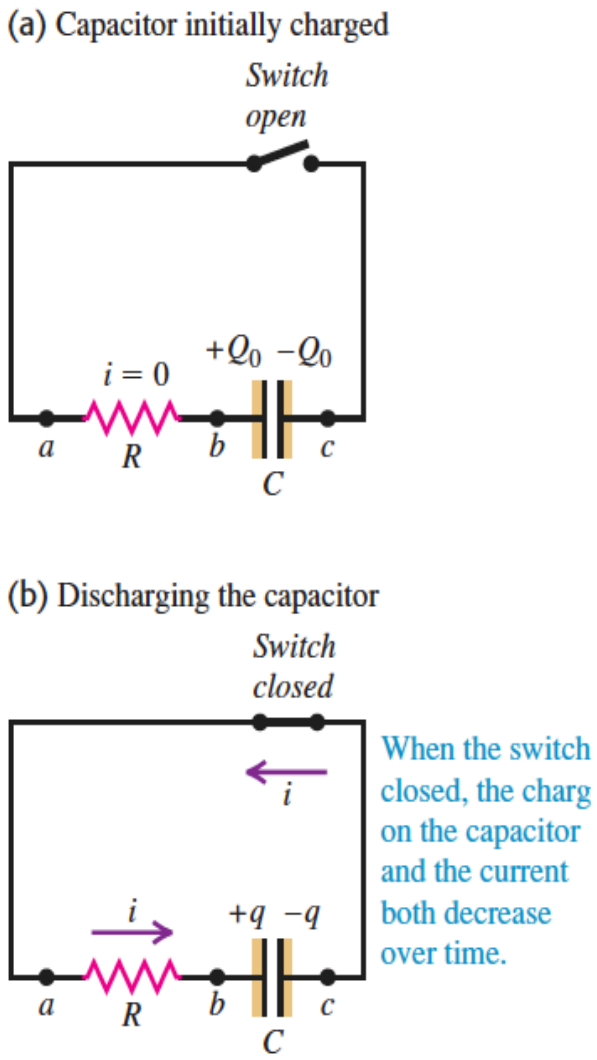


Figure 22 a) Capacitor initially charged b) Discharging the Capacitor. From. Real Source of emf and a bulb.. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 867) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

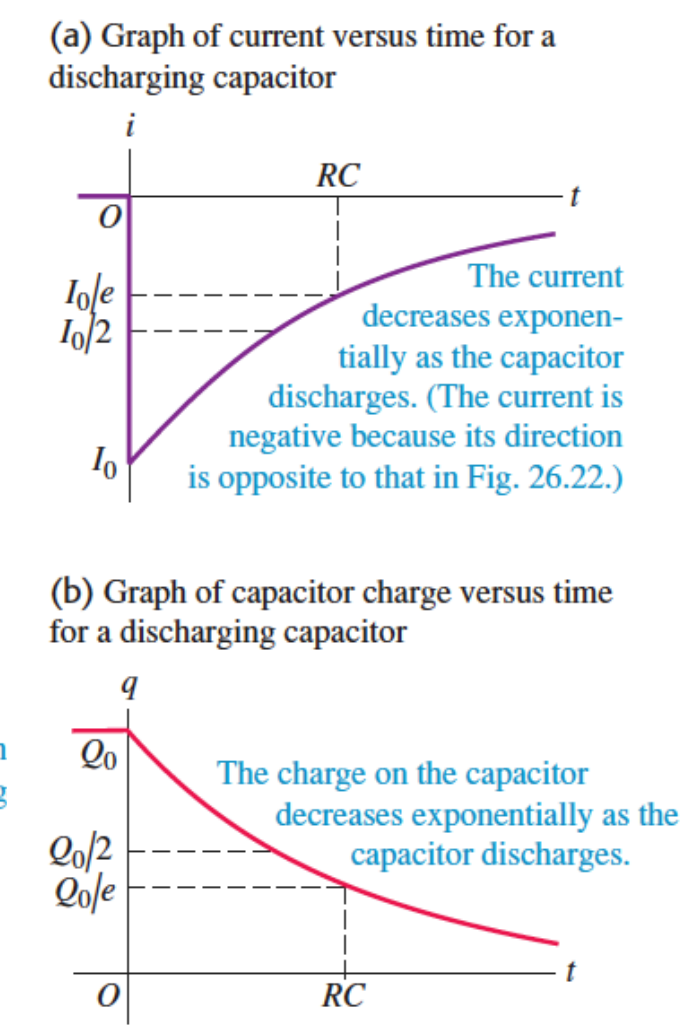


Figure 23 a) Graph of current versus time. b) Graph of capacitor charge versus time.Physics With Modern Physics (pp 865) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson

RC CIRCUITS

DISCHARGING OF THE CAPACITOR

$$V_C = V_0(e^{-t/RC})$$
$$V_C = V_0(e^{-t/\tau})$$

Where V_0 =initial voltage across capacitor, V
R= resistance, Ω
C = capacitance, F
e = 2.718
t = time, s
$\boldsymbol{\tau = RC}$= time constant , s (the time it takes for the voltage across the capacitor to drop to 37% of its initial value)

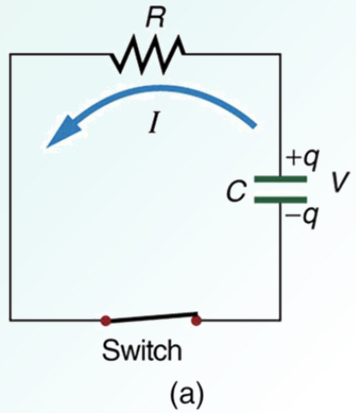
$$Q = Q_0\left(e^{-\frac{t}{RC}}\right)$$
$$Q = Q_0(e^{-t/\tau})$$

Where Q_0 = maximum charge, C
R = resistance, Ω
C = capacitance, F
e = 2.718
t = time, s
$\boldsymbol{\tau = RC}$= time constant , s

Sample Problem

If a charged capacitor, C=35μF is connected to a resistance R = 120 Ω , how much time will elapse until the voltage falls to 10% of its original (maximum) value?

$$t = RCln\left(\frac{V_0}{V_C}\right)$$



$$\tau = RC$$
$$\tau = (120\Omega)(35x10^{-6}F)$$
$$\tau = 4.2x10^{-3}s$$

$$\boldsymbol{V_C = V_0(e^{-t/RC})}$$
$$\boldsymbol{0.10V_0 = V_0(e^{-t/RC})}$$
$$\boldsymbol{e^{-\frac{t}{RC}} = 0.10}$$
$$\boldsymbol{ln\left(e^{-\frac{t}{RC}}\right) = -\frac{t}{RC} = ln(0.10) = -2.3}$$
$$\boldsymbol{t = 2.3(RC)}$$
$$\boldsymbol{t = 2.3(4.2X10^{-3}s)}$$
$$\boldsymbol{t = 9.7x10^{-3}s\ or\ 9.7\ ms}$$

REFERENCES

Serway, R. A., & Jewett, J. W. (2019). Physics for Scientists and Engineers with Modern Physics Tenth Edition. Australia: Cengage Learning.

Young, H. D., & Freedman, R. A. (2016). Sear's & Zemansky's University Physics With Modern Physics. England: Pearson.

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